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Nonclinical Summary

	Document Number: U11-1641-01 **Document ID: EU**
Drug Substance(s):	Linagliptin / Metformin
Dosage Form, Strength:	Linagliptin (L) / Metformin Hydrochloride (M) Film-coated fixed-dose combination (FDC) tablets at dosages of: 2.5 mg / 500 mg 2.5 mg / 850 mg 2.5 mg / 1000 mg
Document Title:	Pharmacokinetics Tabulated Summary
Date of Report:	16 May 2011
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1. PHARMACOKINETICS: OVERVIEW

Test Article: Linagliptin

Type of Study	Study Category	Test System	Method of Administration	Testing Facility	Report Number	Location in Module
Analytical methods and validation reports	Method validation (Quantification of linagliptin)	Mouse and rabbit plasma	NA	BI	U05-1743	4.2.2.1
	Method validation (Quantification of linagliptin and CD 1790)	Mouse plasma	NA	BI	U08-1939	4.2.2.1
	Method validation (Quantification of linagliptin)	Rat, dog and minipigs plasma	NA	BI	U04-1551	4.2.2.1
	Method validation (Quantification of linagliptin)	Rat plasma	NA	BI	U09-2433	4.2.2.1
	Method validation (Quantification of linagliptin and CD 1790)	Rat plasma	NA	BI	U07-1768	4.2.2.1
	Method validation (Quantification of linagliptin and CD 1790)	Rabbit plasma	NA	BI	U08-2118	4.2.2.1
	Method validation (Quantification of linagliptin and CD 1790)	Rabbit plasma	NA	BI	U10-1503	4.2.2.1
	Method validation (Quantification of linagliptin)	Cynomolgus monkey plasma	NA	BI	U04-1807	4.2.2.1
	Method validation (Quantification of linagliptin and CD 1790)	Cynomolgus monkey plasma	NA	BI	U07-2397	4.2.2.1
	Method validation (Quantification of CD 1750, CD 1789, CD 1790)	Human plasma	NA	BI	U08-1831	5.3.1.4

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1. Pharmacokinetics: Overview (Continued) Test Article: Linagliptin / metformin

Analytical methods and validation reports (continued)	Method validation (Quantification of metformin)	Rat plasma	NA	SGS Cephac Europe	U09-2204	4.2.2.1
	Radiosynthesis of [^{14}C]linagliptin	NA	NA	BI	U05-1174	4.2.2.1
Absorption	PK of linagliptin in plasma	Mouse	oral	BI	U06-1746	4.2.2.2
	Influence of P-gp inhibition on absorption	Rat	oral	BI	U09-2236	4.2.2.2
	PK of linagliptin and radioactivity	Rabbit (female)	oral	BI	U08-2294	4.2.2.2
	Comparison of formulations	Cynomolgus monkey	oral	BI	U05-1945	4.2.2.2
	PK of linagliptin and radioactivity in plasma	Cynomolgus monkey	oral & intravenous	MDS France	U06-1132	4.2.2.2
	PK of linagliptin in plasma	Cynomolgus monkey	oral & intravenous	BI	U05-1341	4.2.2.2
Distribution	Plasma protein binding and blood cell partitioning	Rat (Wistar), Dog (Beagle), Minipig (Göttinger) and human	NA (in vitro)	BI	U04-1420	5.3.2.1
	Plasma protein binding	Rat (WT & DPP-4 deficient Fischer) Mouse (WT and DPP-4 knockout) human	NA (in vitro)	BI	U08-1216	5.3.2.1
	Plasma protein binding	Rat (Wistar , WT & DPP-4 deficient Fischer)	NA (in vitro)	BI	U06-2155	4.2.2.3
	Plasma protein binding	Mouse Rabbit (female) Cynomolgus monkey	NA (in vitro)	BI	U05-2596	4.2.2.3

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1. Pharmacokinetics: Overview (Continued) Test Article: Linagliptin

Type of Study	Study Category	Test System	Method of Administration	Testing Facility	Report Number	Location in Module
Distribution (continued)	Plasma protein binding of CD 1790	Rat Cynomolgus monkey Human	NA (in vitro)	BI	U09-1951	5.3.2.1
	Plasma protein binding	Human (renal & hepatic impairment)	NA (in vitro)	Covance, UK	U09-2250	5.3.2.1
	Blood cell partitioning	Human	NA (in vitro)	BI	U09-2476	5.3.2.1
	Blood cell partitioning	Rat, Cynomolgus monkey	NA (in vitro)	BI	U10-1144	4.2.2.3
	Transporter	Pig kidney epithelial cell line	NA (in vitro)	BI (Japan)	U06-3418	4.2.2.5
	Transporter	Human Embryonic Kidney (HEK293) cells LLC-PK1 cells, MDCKII cells	NA (in vitro)	BI (Japan)	U06-3019	4.2.2.3
	Transporter	Human (Caco-2 cells)	NA (in vitro)	Across barriers, Germany	U05-1795	4.2.2.2
	Quantitative whole body autoradiography	Rat (WT & DPP-4 deficient Fischer)	oral & intravenous	BI	U07-1488	4.2.2.3
	Quantitative whole body autoradiography	Rat (Wistar)	intravenous	BI	U07-1985	4.2.2.3
	Quantitative whole body autoradiography	Rat (Wistar & pigmented)	oral & intravenous	BI	U05-1210	4.2.2.3
	Quantitative whole body autoradiography	Rat (pregnant)	oral	BI	U09-2235	4.2.2.3
	Microautoradiography	Rat (Wistar)	intravenous	BI	U08-1913	4.2.2.3
	Tissue distribution	Rat (Wistar)	oral	BI	U05-2059	4.2.2.3
	Tissue distribution after repeated administration	Rat (Wistar)	oral	BI	U05-2013	4.2.2.3
	Tissue distribution after repeated administration	Rat (Wistar)	oral	RCC, Switzerland	U08-1540	4.2.2.3

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1. Pharmacokinetics: Overview (Continued) Test Article: Linagliptin

Type of Study	Study Category	Test System	Method of Administration	Testing Facility	Report Number	Location in Module
Distribution (continued)	Tissue distribution	Mouse (WT and DPP-4 knockout)	intravenous	BI	U08-1364	4.2.2.3
	Tissue distribution	Mouse (WT and DPP-4 knockout) human	intravenous	BI	U08-2214	4.2.2.3
	Tissue distribution	Rat (WT & DPP-4 deficient Fischer)	intravenous	BI	U08-1240	4.2.2.3
	Kidney and liver concentrations of linagliptin	Cynomolgus monkey	oral	BI	U08-2101	4.2.2.3
	PK model and tissue distribution	Rat (WT & DPP-4 deficient Fischer)	NA (modelling)	BI	U09-1028	4.2.2.3
	Distribution into embryos	rats	oral	BI	U10-1332	4.2.2.3
	Distribution into fetus	rabbits	oral	Harlan (Switzerland)	U10-1684	4.2.2.3
Metabolism	In vitro metabolism	Rat Human	NA (in vitro)	BI	U05-2525	4.2.2.4
	Non-extractable radioactivity from plasma	Rat Mouse Rabbit Cynomolgus monkey Human	NA (summary report)	BI	U08-1228	5.3.2.2
	Enzyme induction	Human (hepatocytes)	NA (in vitro)	BD Biosciences Discovery Labware, USA	U08-1198	5.3.2.2
	In vivo metabolism (enantioselectivity of CD 1790 formation)	Human	oral	BI	U07-2471	5.3.1.4
	CYP inhibition	Human (microsomes)	NA (in vitro)	BI	U05-1012	4.2.2.4
	CYP inhibition by CD 1790	Human (microsomes)	NA (in vitro)	BI	U09-1515	4.2.2.4

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1. Pharmacokinetics: Overview (Continued) Test Article: Linagliptin / metformin

Type of Study	Study Category	Test System	Method of Administration	Testing Facility	Report Number	Location in Module
Metabolism (continued)	In vitro metabolism (mechanism of CD 1790 formation)	Rat, Human (liver microsomes, cytosol)	NA (in vitro)	BI	U08-2140	4.2.2.4
	In vivo metabolism	Mouse	oral	BI	U05-2537	4.2.2.4
	In vivo metabolism	Rat	oral & intravenous	BI	U05-1981	4.2.2.4
	In vivo metabolism	Rabbit (female)	oral	BI	U06-2216	4.2.2.4
	In vivo metabolism	Cynomolgus monkey	oral & intravenous	BI	U06-2053	4.2.2.4
	In vivo metabolism	Human	oral & intravenous	BI	U07-1751	5.3.2.2
	In vivo metabolism	Human	oral	BI	U05-2659	5.3.2.2
	In vivo enzyme (CYP) induction	Rat	oral	Quintiles, UK	U04-2193	4.2.2.4
	CYP inhibition by metformin	Human liver microsomes	in vitro	BI	U08-2174	4.2.2.4
Excretion	Excretion balance	Mouse	oral	BI	U05-2607	4.2.2.5
	Excretion balance	Rabbit (female)	oral	BI	U06-1087	4.2.2.5
	Excretion balance	Rat	oral & intravenous	BI	U04-1731	4.2.2.5
	Milk excretion	Rat (lactating)	oral	BI	U08-1929	4.2.2.5
	Excretion balance	Cynomolgus monkey	oral & intravenous	MDS France	U05-2668	4.2.2.5
	Biliary excretion and enterohepatic recirculation	Rat	intraduodenal	BI	U09-1831	4.2.2.5
	Influence of p-gp inhibition on biliary excretion	Rat	intravenous	BI	U09-2339	4.2.2.5
	Dose-dependency of urinary excretion	Rat	oral	BI	U09-2475	4.2.2.5
	Biliary excretion and direct excretion into the gut	Rat	intravenous	BI	U10-2457	4.2.2.5

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1. Pharmacokinetics: Overview (Continued) Test Article: Linagliptin

Type of Study	Study Category	Test System	Method of Administration	Testing Facility	Report Number	Location in Module
Pharmacokinetic Drug Interactions	NA					
Other Pharmacokinetic studies	NA					

NA = not applicable
BI = Boehringer Ingelheim

Note: All studies were performed non-GLP except from study No. C73207 (U10-1684).

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3.A PHARMACOKINETICS: ABSORPTION AFTER A SINGLE DOSE

Test Article: Linagliptin

(Pharmacokinetics of linagliptin after single oral administration to mice)

Location in Module 4: 4.2.2.2
Report No.: U06-1746
Study No.: A049/06FU

Species	Mouse
Strain	CD-1
Gender (M/F) / Number of animals	M / 30
Feeding condition	ad libitum fed
Vehicle/Formulation	solution in tap water
Method of Administration	oral (gavage)
Dose	5, 15 mg/kg
Sample	plasma
Analyte	linagliptin
Assay	HPLC-MS/MS

PK parameters (based on composite data):

		linagliptin dose	
Parameter	**Unit**	**5 mg/kg**	**15 mg/kg**
C_{max}	**[nmol/L]**	84.3	1010
t_{max}	**[h]**	1.00	1.00
AUC_{0-inf}	**[(nmol·h)/L]**	422	3050
$t_{1/2}$	**[h]**	23.6	18.7
MRT_{tot}	**[h]**	11.0	4.99

Additional Information:
A more than proportional increase of AUC_{0-inf} and C_{max} with dose was observed between 5 and 15 mg/kg in male mice, suggesting non-linear mechanisms in the oral pharmacokinetics of linagliptin. Long terminal half-lives and moderate to long MRTs of linagliptin were observed. The interindividual variability of the linagliptin plasma levels was moderate to high.

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3.B PHARMACOKINETICS: ABSORPTION AFTER A SINGLE DOSE

Test Article: Linagliptin

(Pharmacokinetics of linagliptin after single oral administration to rats)

Location in Module 4: 4.2.2.5
Report No.: U04-1731
Study No.: A017/04FU

Species	Rat
Strain	Crl:WI(Han)
Gender (M/F) / Number of animals	M / 4 per dose group
Feeding condition	oral: fasted i.v.: ad libitum fed
Vehicle/Formulation	solution
Method of Administration	oral (gavage) i.v. (bolus injection)
Dose	5, 15, 45 mg/kg linagliptin 1 mg/kg [^{14}C]linagliptin
Sample	plasma
Analyte	linagliptin, radioactivity
Assay	HPLC-MS/MS, LSC

PK parameters of linagliptin:

Parameter	Unit	5 mg/kg i.v.	5 mg/kg p.o.#	15 mg/kg p.o.#	45 mg/kg p.o.#
C_{max}	[nmol/L]	5330	547	2230	6830
t_{max}*	[h]	0.033	2.25	0.5	1.25
AUC_{0-inf}	[nmol·h/L]	4760	2610	8360	45600
$t_{1/2}$	[h]	10.5	35.9	29.8	12.2
MRT	[h]	2.40	14.3	6.81	5.51
CL	[(mL/min)·kg)]	37.3	NA	NA	NA
V_{ss}	[L/kg]	5.39	NA	NA	NA
MAT	[h]	NA	11.9	4.41	3.11

NA = not applicable
* = median, # = fasted rats

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3.B Pharmacokinetics: Absorption after a Single Dose (continued)

Report No.: U04-1731

PK parameters of radioactivity:

Parameter	Unit	1 mg/kg i.v.	1 mg/kg p.o.#
C_{max}	[nmol/kg]	1330	61.9
t_{max}*	[h]	0.033	4
$AUC_{0\text{-}inf}$	[nmol·h/kg]	1860	695
$t_{1/2}$	[h]	36.4	36.5
MRT	[h]	27.2	39.8

* = median, # = fasted rats

Additional Information:
Linagliptin is well absorbed by rats as indicated by an oral bioavailability of 50.7%. At 5 mg/kg linagliptin, the compound showed a delayed absorption in fasted rats, coinciding with the time of feeding. This resulted in a biphasic time course of the linagliptin plasma levels at this dose. The delayed absorption is also reflected by a long MAT of 11.9 hours. In comparison, the AUC values calculated from blood radioactivity data were similar in fasted and ad libitum fed rats. At higher doses administered orally, the MAT was decreased (4.41 or 2.11 at 15 or 45 mg/kg linagliptin, respectively). Taken together, a moderate to slow rate of absorption was observed.

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3.C PHARMACOKINETICS: ABSORPTION AFTER A SINGLE DOSE

Test Article: Linagliptin

(Influence of intestinal P-gp on oral bioavailability of linagliptin in rats)

Location in Module 4: 4.2.2.2
Report No.: U09-2236
Study No.: A056/08HH

Species	Rat / Wistar			
Strain	Crl:WI(Han)			
Gender (M/F) / Number of animals	M / 4 per group			
Feeding condition	fasted			
Formulation	solution			
Method of Administration	oral (gavage)			
Dose linagliptin	1 mg/kg	1 mg/kg	15 mg/kg	15 mg/kg
Pre-treatment Zosuquidar (p.o., - 5 min)	0 µmol/kg	5 µmol/kg	0 µmol/kg	5 µmol/kg
Sample	plasma			
Analytes	linagliptin CD 1790			
Assay	HPLC-MS/MS			

dose-normalized PK parameters linagliptin:					
C_{max}/dose	[(nmol/L)/(µmol/kg)]	12.0	164	88.7	120
AUC_{0-72h}/dose	[(nmol·h/L)/(µmol/kg)]	147	309	227	309
$t_{1/2}$	[h]	39.8	48.6	36.7	34.8
MRT_{tot}	[h]	48.2	32.3	6.34	7.04
dose-normalized PK parameters CD 1790:					
C_{max}/dose	[(nmol/L)/(µmol/kg)]	0.438	3.24	3.59	4.95
AUC_{0-72h}/dose	[(nmol·h/L)/(µmol/kg)]	0.969	5.66	10.5	18.2
Ratios CD 1790 / linagliptin for:					
C_{max}	[%]	3.5	2.6	4.1	4.5
AUC_{0-72h}	[%]	0.6	2.0	4.6	5.8

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3.C Pharmacokinetics: Absorption after a Single Dose (continued)

Report No.: U09-2236

Additional Information:
The study was performed in order to investigate the influence of intestinal P-glycoprotein (P-gp) on the oral pharmacokinetics of linagliptin in the rat using the P-gp inhibitor Zosuquidar. Over-proportional increase of C_{max} and AUC_{0-72h} was observed between 1 and 15 mg/kg linagliptin. Pre-treatment with Zosuquidar resulted in a distinct increase of C_{max} and AUC_{0-72h}, however, the extent of the Zosuquidar-effect was dependent on the oral dose of linagliptin: C_{max} was elevated by factors of approximately 14 and 1.4 at oral linagliptin doses of 1 and 15 mg/kg, respectively. The elevation factor of AUC_{0-72h} was 2.1 and 1.4, respectively. Low plasma concentrations of CD 1790 in all groups precluded complete PK evaluation for this analyte. However, the ratio CD 1790 over linagliptin for C_{max} and AUC_{0-72h} was not relevantly changed by pre-treatment with Zosuquidar, arguing against inhibition of CYP 3A by the Zosuquidar pre-treatment. In conclusion, oral absorption of linagliptin is enhanced by inhibition of intestinal P-gp, resulting in a substantial increase in C_{max} and AUC_{0-72h} in the rat. Auto-inhibition of intestinal P-gp by high doses of linagliptin seems to be the cause for the over-proportional increase in C_{max} And AUC_{0-72h} between 1 and 15 mg/kg linagliptin p.o.

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3.D PHARMACOKINETICS: ABSORPTION AFTER A SINGLE DOSE

Test Article: Linagliptin

(Pharmacokinetics of linagliptin after single oral administration to female rabbits)

Location in Module 4: **4.2.2.2**
Report No.: **U08-2294**
Study No.: **A024/07JS**

Species	Rabbit	
Strain	Himalayan CHBB(HM)	
Gender (M/F) / Number of animals	F / 3 per dose group	
Feeding condition	fed	
Vehicle/Formulation	solution	
Method of administration	oral (gavage)	
Dose	4 and 25 mg/kg	
Sample	plasma	
Analyte	linagliptin	CD 1790
Assay	HPLC-MS/MS	HPLC-MS/MS

	4 mg/kg		**25 mg/kg**	
Pharmacokinetic parameter [units]	linagliptin	CD 1790	linagliptin	CD 1790
C_{max} [nM]	368	19.3	8560	307
t_{max} [h]	0.67	0.83	0.50	0.67
AUC_{0-96h} [(nmol·h)/L]	1330	38.1	16300	650
$t_{1/2}$ [h]	41.3	NC	59.0	15.5
MRT_{tot} [h]	33.9	NC	9.98	8.13

NC = not calculated

Additional Information:
A more than proportional increase of AUC_{0-96h} and C_{max} to dose was seen following single oral administration of 4 and 25 mg/kg linagliptin to female rabbits, suggesting non-linear mechanisms in the oral pharmacokinetics of linagliptin and its metabolite CD 1790. The observed terminal half-lives ($t_{1/2}$) of parent compound and metabolite were long and in the same order of magnitude for both doses, whereas the mean residence time (MRT_{tot}) of linagliptin decreased from about 34 to about 10 hours, supporting the suggested non-linearity in the pharmacokinetics of linagliptin in rabbits.

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3.E PHARMACOKINETICS: ABSORPTION AFTER A SINGLE DOSE

Test Article: Linagliptin

(Pharmacokinetics of linagliptin after single oral administration to cynomolgus monkey)

Location in Module 4: **4.2.2.2**
Report No. **U05-1341**
Study No.: **A020/04FU**

Species	Cynomolgus monkey
Gender (M/F) / Number of animals	M / 4
Feeding condition	fasted 12 h prior to administration, food 4 h after dosing
Vehicle/Formulation	oral: solution in tap water i.v.: solution in saline
Method of Administration	oral (gavage) i.v. (injection into arm vein)
Dose (mg/kg)	3 mg/kg
Sample	plasma
Analyte	linagliptin
Assay	HPLC-MS/MS

PK parameters:

Parameter	Dose / Route [Unit]	3 mg/kg i.v.	3 mg/kg p.o.
C_{max}	[nmol/L]	3000	622
t_{max}*	[h]	0.083 - 0.117	0.5 - 2
AUC_{0-inf}	[nmol·h/L]	5440	3740
$t_{1/2}$	[h]	25.5	46.7
MRT_{tot}	[h]	7.93	28.9
CL	[mL/min/kg]	19.7	NA
V_{ss}	[L/kg]	9.47	NA
MAT	[h]	NA	21.0
F	(%)	NA	68.8

*data given as range
NA = not applicable

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3.F PHARMACOKINETICS: ABSORPTION AFTER A SINGLE DOSE

Test Article: Linagliptin

(Pharmacokinetics of linagliptin after single oral administration to cynomolgus monkey – comparison of two formulations)

Location in Module 4: **4.2.2.2**
Report No.: **U05-1945**
Study No.: **A025/04FU**

Species	Cynomolgus monkey
Gender (M/F) / Number of animals	M / 2 & F / 2
Feeding condition	fasted 12 h prior to administration, food 4 h after dosing
Vehicle/Formulation	solution in tap water or tartaric acid
Method of Administration	oral (gavage)
Dose	3 mg/kg
Sample	plasma
Analyte	linagliptin
Assay	HPLC-MS/MS

PK parameters:

Parameter	Formulation Unit	tap water mean (CV)	tartaric acid mean (CV)
C_{max}	[nmol/L]	700 (18.7%)	699 (21.0%)
t_{max}*	[h]	0.5 (0.5 - 2)	0.75 (0.5 - 2)
$AUC_{0\text{-}inf}$	[nmol·h/L]	4140 (11.5%)	3320 (4.9%)
$t_{1/2}$	[h]	39.7 (3.3%)	36.4 (5.9%)
MRT_{tot}	[h]	19.2 (30.2%)	16.1 (9.8%)

*data given as median and range

Both formulations (tap water or tartaric acid) produced very similar plasma level-time profiles of linagliptin. In addition, the pharmacokinetic parameters of both groups were very similar. Only slight differences were observed regarding AUC, where the tap water formulation produced slightly (about 25%) higher values than the tartaric acid formulation. However, interindividual variability may also account for this. In conclusion, no clear-cut difference between the pharmacokinetics of a tap water or tartaric acid formulation of linagliptin in cynomolgus monkeys could be shown in this study.

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4. PHARMACOKINETICS: ABSORPTION AFTER REPEATED DOSES

Test Article: Linagliptin

Tissue distribution studies including excretion data in one study were performed in rats also after repeated oral dosing. The results are given in 2.6.5.5.

In addition, toxicokinetic data after repeated oral dosing are available from mice, rats, rabbits, dogs and cynomolgus monkeys and toxicokinetic data after repeated short term intravenous infusion are available in rats and cynomolgus monkeys. The data are discussed in the toxicology section (2.6.6.7).

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5.A PHARMACOKINETICS: ORGAN DISTRIBUTION

Test Article: Linagliptin

Location in Module 4: 4.2.2.3
Report No.: U08-1364
Study No.: A084/07FU

Species: Mouse
Strain: DPP-4 knockout (C57BL/6TgH(CD26)-CIML)
wildtype (C57BL/6J)
Gender (M/F)/Number of animals: F / 36
Feeding condition: ad libitum
Vehicle/Formulation: solution
Method of Administration: intravenous bolus injection
Dose (mg/kg): 0.01 - 10
Radionuclide: ^{14}C
Specific Activity (MBq/µmol): 2.14 - 0.21
Sampling time: 72 h

Tissues/organs	**Radioactivity concentration [nmol/kg]**								
dose (mg/kg)	Liver	Duodenum	Carcass	Skin	Lungs	Spleen	Kidneys	Thymus	Sceletal Muscle
	DPP-4 knockout mice								
0.01	2.59	BLQ	BLQ	1.03	BLQ	BLQ	BLQ	BLQ	BLQ
0.1	7.38	1.20	2.03	2.24	NC	NC	1.19	BLQ	BLQ
0.3	18.9	1.33	3.59	1.65	1.80	4.56	3.29	5.65	BLQ
1	70.5	5.17	20.9	6.68	2.91	8.04	10.2	6.55	0.764
3	176	7.73	26.2	13.5	10.7	17.9	26.4	16.7	2.23
10	580	22.4	118	36.5	43.5	83.6	145	62.8	7.04
	Wildtype mice								
0.01	131	11.9	5.68	9.44	37.3	13.4	215	14.7	1.32
0.1	546	102	28.1	52.8	131	50.4	474	111	4.12
0.3	638	92.4	35.8	78.3	145	55.1	533	133	4.43
1	652	133	48.2	77.4	113	47.7	477	129	4.80
3	716	162	66.4	105	176	97.8	517	150	7.99
10	1350	190	171.5	130	273	NOR	687	240	15.5

BLQ = below limit of quantification
NC = not calculated
NOR = no valid result

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5.A Pharmacokinetics: Organ Distribution (Continued)

Report No.: U08-1364

Additional information:

Prominent differences were observed in the tissue distribution of [^{14}C]linagliptin related radioactivity. Radioactivity concentrations in the investigated tissues were markedly lower in the DPP-4 knockout mice at all dose levels. In wildtype mice, the highest concentration of DPP-4 binding sites is suggested in liver followed by the kidney. The gut (duodenum) and the skin contribute also to the total pool of DPP-4 in the mouse body to a relevant extent, despite markedly lower concentrations. The data indicate, that DPP-4 specific binding of linagliptin is saturated already at very low intravascular doses between 0.01 and 0.1 mg/kg linagliptin in wildtype mice. Thus, after repeated administration, accumulation of linagliptin in tissues is unlikely despite its known long residence time in the respective tissue after single dose administration. Furthermore, it is concluded that binding of linagliptin to DPP-4 in tissue is reversible and of high affinity.

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5.B PHARMACOKINETICS: ORGAN DISTRIBUTION

Test Article: Linagliptin

Location in Module 4: 4.2.2.3
Report No.: U05-1210
Study No.: A120/04RB, A127/04RB

Species: Mouse
Strain: CD-1
Gender (M/F)/Number of animals: M / 3
Feeding condition: fasted
Vehicle/Formulation: aqueous solution acidified with HCl
Method of Administration: oral (gavage)
Dose: 2 mg/kg
Radionuclide: ^{14}C
Specific Activity: 1.935 MBq/µmol
Assays: Whole Body Autoradiography

Mean concentration of radioactivity:	[nmol/kg tissue]		
Sampling time:	**24 h**	**72 h**	**168 h**
Harder's gland	53	47	25.3
brain	BLQ	BLQ	BLQ
salivary gland	160	94.9	63.1
thymus	100	72.4	37.8
lungs	76.7	63.6	36.1
liver	377	292	160
spleen	54.9	42.3	26.7
kidney, medulla	48.9	52.5	24.3
kidney, zona intermedia	398	333	209
kidney, cortex	289	250	134
kidney	251	238	132
muscle	BLQ	BLQ	BLQ
epididymus	47.9	68.5	72.6
skin (abdom.)	52.2	31.2	BLQ
skin (back)	62.4	41.2	27.6
Glandula vesicularis	84.3	86	82.7

BLQ = below limit of quantification

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5.B Pharmacokinetics: Organ Distribution (Continued)

Report No.: **U05-1210**

Additional information:	In mice the distribution pattern was qualitatively comparable to that in rats. The retention in the zona intermedia of the kidney, however, was much less pronounced than in the rats. The overall concentration level in mice was lower compared to rats

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5.C PHARMACOKINETICS: ORGAN DISTRIBUTION

Test Article: Linagliptin

Location in Module 4: 4.2.2.3
Report No.: U08-2214
Study No.: A087/07FU

Species: Mouse
Strain: DPP-4 knockout (C57BL/6TgH(CD26)-CIML)
wildtype (C57BL/6J)
Gender (M/F) / Number of animals: M / 13 & F / 14 (DPP-4 knockout), M / 14 & F / 14 (WT)
Feeding condition: ad libitum
Vehicle/Formulation: solution in NaCl
Method of Administration: i.v. bolus injection
Dose: 1 mg/kg
Radionuclide: ^{14}C
Specific Activity: 2.138 MBq/µmol
Sampling time: once daily for 14 days
Study design:
Aim of the present study was to investigate the time course of radioactivity concentrations in kidney, liver, lungs and the skin after a single intravenous bolus administration of 2.12 µmol/kg (1 mg/kg) [^{14}C]linagliptin to either DPP-4 knockout (DPP-4 ko) or wildtype (WT) mice. The tissue concentrations were investigated at 0.5, 3, 24, 48, 72, 168, 336, and 672 h (28 days) after dosing. In addition, selected tissue samples of kidney and liver taken at 0.5, 25 and 168 h after dosing were investigated in detail using an extraction procedure in order to determine the extractable fraction.

Pharmacokinetic parameters of radioactivity in kidney, liver, skin and lungs after a single intravenous (bolus) administration of 2.12 µmol/kg [^{14}C]linagliptin to DPP-4 ko and WT mice.

		Kidney		Liver		Skin		Lungs	
Parameter	**Unit**	DPP-4 ko	WT	DPP-4 ko	WT	DPP-4 ko	WT	DPP-4 ko	WT
C_{max}	**[nmol/L]**	5000	4870	5420	5840	584	554	1510	1910
AUC_{0-inf}	**[(nmol·h)/L]**	22600	212000	93100	256000	7100	24300	9910	59000
$t_{1/2}$	**[h]**	282	374	405	319	253	253	215	297

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5.C Pharmacokinetics: Organ Distribution (Continued)

Report No.: U08-2214

Additional Information:

Taken together, markedly higher exposure of radioactivity was found in tissues of wildtype mice than in DPP-4 knockout mice. C_{max}, which was observed at the first sampling time point (0.5 h), was similar in both strains but marked differences in the concentrations were obvious from the second sampling occasion (3 h) onwards. Thus, the data clearly indicate that persistent, i.e. high affinity binding of linagliptin to its target DPP-4 in tissues predominates its tissue distribution early after dosing.

Gender differences were suggested in the radioactivity concentrations in the kidneys of wildtype mice, which may indicate a gender difference in the DPP-4 concentration in kidneys. In the remaining investigated tissues, no relevant gender differences were observed.

The terminal half-life of radioactivity in the investigated tissues were very long and accounted for >200 h in organs of both, wildtype and DPP-4 knockout mice. It was shown that a very small fraction of the respective sample radioactivity at later time points after dosing was non-extractable. In relation to the administered dose, the non-extractable part was very low and therefore regarded as negligible.

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5.D PHARMACOKINETICS: ORGAN DISTRIBUTION

Test Article: Linagliptin

Location in Module 4: 4.2.2.3
Report No.: U05-1210
Studies No.: A120/04RB, A127/04RB

Species:	Rat (albino)		Rat (pigmented)
Strain:	Wistar Crl:WI(Han)		BN/Crl Br
Gender (M/F)/Number of animals:	M / 4	M / 4	M / 2
Feeding condition:	fasted	fasted	fasted
Vehicle/Formulation:	aqueous solution acidified with HCl	aqueous solution acidified with HCl	solution acidified with HCl
Method of Administration:	oral (gavage)	intravenous	intravenous
Dose:	2 mg/kg	2 mg/kg	2 mg/kg
Radionuclide:	^{14}C		
Specific Activity:	1.13 MBq/µmol		
Assays:	Whole Body Autoradiography		

Mean concentration of radioactivity:	[nmol/kg tissue]				[nmol/kg tissue]				[nmol/kg tissue]	
Sampling time:	**0.5 h**	**4 h**	**24 h**	**168 h**	**2 min**	**3 h**	**24 h**	**168 h**	**3 h**	**168 h**
eye									10600	6460
Harder's gland	514	232	84.7	92.6	4470	3960	157	ND	5000	317
tongue	516	146	BLQ	BLQ	6300	542	BLQ	BLQ	559	BLQ
brain	BLQ	BLQ	BLQ	BLQ	36.9	BLQ	BLQ	BLQ	BLQ	BLQ
pituitary	1100	329	57.9	ND	ND	3400	ND	ND	4870	ND
spinal cord	BLQ	BLQ	BLQ	BLQ	73	BLQ	BLQ	BLQ	BLQ	BLQ
brown fat*	613	187	38.6	BLQ	5360	989	42.6	ND	684	BLQ
salivary gland	1230	461	330	52.8	12000	3190	308	47.5	2910	67.8
thymus	290	151	299	111	2030	1520	344	125	1590	143
heart muscle	603	199	73.2	BLQ	14800	504	89.3	BLQ	554	BLQ
blood (heart)	144	41.6	BLQ	BLQ	2880	140	BLQ	BLQ	126	BLQ
lungs	706	461	344	86.6	8120	1680	319	58	1100	59.5
liver	5650	1760	1250	304	12100	4450	1610	447	3340	333
spleen	1930	1090	671	135	4660	2560	550	173	3410	195
adrenal	1730	563	270	45.8	26600	1730	243	78.4	2310	88.9

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5.D Pharmacokinetics: Organ Distribution (Continued)

Report No.: U05-1210

Mean concentration of radioactivity:	[nmol/kg tissue]				[nmol/kg tissue]				[nmol/kg tissue]	
Sampling time:	**0.5 h**	**4 h**	**24 h**	**168 h**	**2 min**	**3 h**	**24 h**	**168 h**	**3 h**	**168 h**
kidney, medulla	2520	670	2630	85.4	12600	1670	291	133	2610	194
kidney, zona intermedia	ND	9970	10600	9160	ND	13600	10100	9240	11000	7510
kidney, cortex	ND	3960	3620	2010	29600	5360	3650	2010	4060	2070
kidney, total	5230	4690	4020	2200	31600	6380	4380	2200	6060	2420
bone marrow	744	278	133	25.7	3490	ND	181	34.2	ND	47.8
muscle	323	87.7	BLQ	BLQ	5970	677	BLQ	BLQ	547	BLQ
testis	ND	ND	29.5	BLQ	171	270	154	ND	260	BLQ
epididymis	ND	ND	251	67.6	1100	868	126	97.2	1090	169
fat*	51.8	11.5	BLQ	BLQ	173	63.7	BLQ	BLQ	47	BLQ
skin (abdom.)	237	123	61.1	BLQ	1270	437	57.7	BLQ	546	87.6
skin (back)	220	153	74.0	27.0	1350	555	106	50.1	543	41

ND = not determined
BLQ = below limit of quantification (27 nmol/kg)
*: potentially underestimated due to elevated self-absorption in adipose tissues

Additional information:	In contrast to the majority of the tissues, a retention of radioactivity was found in the kidney of the rat. A pronounced blood brain barrier was also seen after oral administration. After intravenous administration marked affinity and retention was observed in the kidney, especially in the zona intermedia. In addition, there was a certain affinity of the radioactivity to the melanin containing part of the eye of pigmented rats

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5.E PHARMACOKINETICS: ORGAN DISTRIBUTION

Test Article: Linagliptin

Location in Module 4: 4.2.2.3
Report No.: U08-1240
Study No.: A057/06FU

Species: Rat
Strain: DPP-4 deficient Fischer rat (F344/DuCrlCrlJ)
wildtype Fischer rat (F344/DuCrl)
Gender (M/F)/Number of animals: M / 72
Feeding condition: ad libitum
Vehicle/Formulation: solution
Method of Administration: intravenous bolus
Dose: 0.01 - 50 mg/kg
Specific Activity: 0.034 - 1.459 MBq/µmol
Radionuclide: ^{14}C
Sampling time: 72 h

Tissues/organs		Concentration [nmol/kg]						
Strain	Dose [mg/kg]	Kidneys	Liver	Lungs	Spleen	Subm. gland	Thymus	Skin
WT	0.01	485	215	123	45.5	39.2	29.1	7.54
DPP-4 deficient	0.1	8.27	33.2	1.26	5.15	2.47	2.92	0.748
WT	0.1	3760	528	390	157	99.9	185	43.6
DPP-4 deficient	0.3	28.2	95.9	3.78	11.4	6.54	3.36	1.4
WT	0.3	3990	608	467	204	127	259	51.3
DPP-4 deficient	1	106	296	14.1	34.4	10.9	9.01	5.24
WT	1	4120	816	525	233	160	251	53.0
DPP-4 deficient [a)]	1	122	307	26.9	135	36.8	24.3	5.58
WT [a)]	1	3380	874	591	399	151	282	43.8
DPP-4 deficient	3	660	368	111	85.6	82.5	96.7	16.6
DPP-4 deficient	10	1180	2220	154	594	264	157	53.8
WT	10	5310	2810	929	862	485	415	115
DPP-4 deficient	50	5030	9290	888	2780	700	605	312
WT	50	9280	8920	1960	3320	1020	918	299

a) older animals

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5.E Pharmacokinetics: Organ Distribution (Continued)

Report No.: U08-1240

Additional information:
Prominent differences were observed in the radioactivity concentrations in the investigated tissues which were markedly lower in the DPP-4 deficient rats at all dose levels. The major pool of DPP-4 is located in special tissues, mainly liver and kidney. The data indicate, that DPP-4 specific binding of linagliptin is saturated already at very low intravascular doses between 0.01 and 0.1 mg/kg linagliptin in wildtype rats. Thus, after repeated administration, accumulation of linagliptin in tissues is unlikely despite its known long residence time in the respective tissue after single dose administration. Furthermore, it is concluded that binding of linagliptin to DPP-4 in tissue is reversible and of high affinity. Data of this study have been published [P09-08326].

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5.F PHARMACOKINETICS: ORGAN DISTRIBUTION

Test Article: Linagliptin

Location in Module 4: 4.2.2.3
Report No.: U05-2013
Study No.: A035/05FU

Species:	Rat
Strain:	Wistar Crl:WI(Han)
Gender (M/F) / Number of animals:	M/3 (1 day), M/4 (14 days), M/4 (21 days)
Feeding condition:	ad libitum
Vehicle/Formulation:	solution in tap water
Method of Administration:	oral (gavage)
Dose:	2 mg/kg
Radionuclide:	^{14}C
Specific Activity:	about 0.07 - 0.17 MBq/µmol
Sampling time:	1, 14, 21 days after dosing

Mean tissue concentrations of [^{14}C]-related linagliptin after repeated once daily oral administration of 2 mg/kg [^{14}C]linagliptin to rats for either 1, 14, or 21 days. Data are expressed as nmol/kg.

	Kidney		**Liver**		**Lungs**		**Spleen**		**Thymus**		**Epididymis**		**Muscle**		**Testes**		**Blood**		**Plasma**	
Day	mean	SD	mean	SD	mean	SD	mean	SD	mean	SD	mean	SD	mean	SD	mean	SD	mean	SD	mean	SD
1	3990	139	1020	296	649	45	490	65	217	100	89.5	41	12.3	2.43	24.7	13	11.6	1.36	17.5	0.61
14	4900	398	1420	131	969	262	637	24	310	39	86	27	15.2	0.8	31.9	10	98.7*	55.9*	19.9	3.38
21	6260	563	1670	252	941	90	746	83	311	15	137	17	13.2	1.49	34.2	2.6	39.6	26.7	11.1	0.85

* excluded

Additional information:

Within the kidney, the highest concentrations of [^{14}C]-related linagliptin were detected in an intermediate zone which was assigned to the outer band of the medulla (7764 nmol/kg, day 1), followed by the renal cortex (4020 nmol/kg, day 1). In contrast, the renal medulla (inner medulla) showed only low concentrations of [^{14}C]-related linagliptin (314 nmol/kg, day 1). In comparison to day 1, [^{14}C]-related linagliptin levels were slightly increased on day 14. However, between day 14 and day 21 no further increase was observed. The accumulation ratios ranged between 1.26 and 1.77.

Based on the findings of the present study in rats, binding of [^{14}C]-related linagliptin to various tissues and in particular within the kidneys after repeated once daily oral administrations of 2 mg/kg linagliptin appears to be in steady state conditions very quickly. In addition, no relevant tissue accumulation of [^{14}C]-related linagliptin was observed after 14 or 21 days of repeated once daily oral treatments with 2 mg/kg linagliptin, making relevant further accumulation after longer treatment periods unlikely.

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5.G PHARMACOKINETICS: ORGAN DISTRIBUTION

Test Article: Linagliptin

Location in Module 4: **4.2.2.3**
Report No.: **U05-2059**
Study No.: **A027/05FU, A440/05BC, A238/05TE**

Species:	Rat (Wistar)
Strain:	Wistar Crl:WI(Han)
Gender (M/F) / Number of animals:	M / 4 per dose group
Feeding condition:	ad libitum
Vehicle/Formulation:	solution in tap water
Method of Administration:	oral (gavage)
Dose:	2 mg/kg (with pre-treatment of non-labelled linagliptin 2 or 60 mg/kg, or no pre-treatment)
Radionuclide:	^{14}C
Specific Activity (MBq/µmol):	about 1
Sampling time:	24 h p.a.

Study design:

Aim of the present study was to investigate the binding of [^{14}C]linagliptin related radioactivity to rat kidney and liver after a single oral administration. In particular, the influence of an oral pre-treatment by non-labelled linagliptin in either an equimolar or excess dose was studied in order to assess the saturability of linagliptin tissue binding. Additionally it was investigated whether [^{14}C]linagliptin related radioactivity bound to tissues could be assigned to parent compound or metabolites and whether binding occurs in a covalent or non-covalent manner. For this purpose, four groups of rats were included in this study. Group 1 served as control group. Animals were neither pre-treated nor treated as only blank tissue samples were collected from these animals. Group 2 was treated with 2 mg/kg [^{14}C]linagliptin without pre-treatment. Group 3 was pre-treated with an equimolar dose of 2 mg/kg linagliptin prior to treatment with 2 mg/kg [^{14}C]linagliptin. Group 4 was pre-treated with an excess dose of 60 mg/kg linagliptin (30 fold higher than treatment) before treatment with 2 mg/kg [^{14}C]linagliptin .

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5.G Pharmacokinetics: Organ Distribution (Continued)

Report No.: U05-2059

	group 2: no pre-treatment		**group 3: pre-treatment with 2 mg/kg linagliptin**		**group 4: pre-treatment with 60 mg/kg linagliptin**	
Sample	Mean linagliptin conc. nmol/kg	extractable fraction %	Mean linagliptin conc. nmol/kg	extractable fraction %	Mean linagliptin conc. nmol/kg	extractable fraction %
Spleen	579	ND	324	ND	64.4	ND
Liver	841	88.6	481.7	88.9	184	63.2
Kidney	4360	100.1	2540	101.2	230	96.8

ND = not determined

Additional information:
A detailed analysis using LC-MS revealed, that that the extractable, non-covalently bound radioactivity in the kidney was entirely due to the parent drug. Liver tissue extracts showed the presence of trace amounts of metabolites of linagliptin, which was to be expected as the liver is the major organ for metabolism and linagliptin can be expected to undergo some degree of metabolism in the rat.

In conclusion, after oral dosing of linagliptin to rats only parent compound is bound non-covalently in rat kidney tissue. The binding to kidney tissue is saturable, indicating specific binding sites with low capacity. In rat liver, a minor fraction of linagliptin binds covalently to liver tissue whereas the remaining major fraction is bound non covalently and consists almost exclusively of parent compound. In summary, the parent drug linagliptin exhibited saturable, non-covalent binding to liver and kidney tissue and the presence of metabolites was not relevant for this observation.

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5.H PHARMACOKINETICS: ORGAN DISTRIBUTION

Test Article: Linagliptin

Location in Module 4: 4.2.2.3
Report No.: U08-1540
Study No.: B84497

Species:	Rat (Wistar)
Strain:	Wistar Crl:WI(Han)
Gender (M/F) / Number of animals:	M / 15 & F / 9
Feeding condition:	ad libitum
Vehicle/Formulation:	solution in tap water
Method of Administration:	oral (gavage)
Dose:	2 mg/kg
Radionuclide:	^{14}C
Specific Activity:	about 0.24 MBq/µmol
Sampling time:	trough (24 h p.a.) values on Day 1, 4, 7, 10 and 14

Study design:

The disposition of linagliptin was investigated in the rats after multiple oral administration. ^{14}C labelled linagliptin was orally administered to two groups of rats consisting of 15 males and 9 females. The dosing was performed by 14 consecutive daily doses at a nominal dose level of 2 mg/kg body weight. The animal groups were further divided into subgroups of 3 animals each, i.e. T1, T2, T3, T4, T5, (males) and T1, T2, T3 (females) and the trough concentration in tissues and organs were determined 24 h after the last dose. Urine and faeces of the Subgroup T5 (males) were collected in daily intervals in order to evaluate the excretion profile during and after a multiple dosing regimen and the recovery of the administered radioactivity (cf. 2.6.5.13/14). Data of this study have been published [P09-08326].

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5.H Pharmacokinetics: Organ Distribution (Continued)

Report No.: U08-1540

	Tissue concentrations of [^{14}C]linagliptin derived radioactivity										
Day	Kidneys	Liver	Lung	Spleen	Thymus	Bone marrow	Epididymis	Lymph-nodes	Adrenal	Submand. gland	Duodenum
1	2650	719	580	417	104	265	39	149	139	142	63
4	5011	879	582	466	315	276	112	183	176	149	121
7	5189	941	609	525	343	257	160	216	194	186	85
10	4920	905	668	506	343	248	178	190	177	152	73
14	4988	944	689	630	341	291	225	213	201	190	133

Day	Skin	Pancreas	Heart	Fat-brown	Testes	Fat-white	Muscle	Thyroid	Pituitary	Plasma	Blood	Brain
1	78	69	49	*30	*12	*21	*11	*56	<LOD	*7	<LOD	<LOD
4	117	102	55	*28	22	*12	*13	*59	<LOD	*9	*9	′=LOD
7	104	100	72	42	25	*22	19	*77	<LOD	*9	*9	*8
10	120	119	62	41	19	*18	*15	*61	<LOD	*10	*10	<LOD
14	106	98	69	42	29	*20	*13	*70	*91	*11	*12	*7

* <LOQ, quantitative data have to be taken with care
LOD = limit of detection

In summary, the trough tissue concentrations increased from Day 1 to Day 4 but did not increase afterwards. Thus, steady state is achieved quickly after the 4th dose at the latest.

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5.I PHARMACOKINETICS: ORGAN DISTRIBUTION

Test Article: Linagliptin

Location in Module 4: 4.2.2.3
Report No.: U07-1488
Study No.: A134/06GR

Species/Strain:	rat / F344 (DPP-4 deficient)	rat / F344 (DPP-4 deficient)	rat / F344 (wildtype)	rat / F344 (wildtype)
Gender (M/F)/Number of animals:	M / 3	M / 2	M / 3	M / 2
Feeding condition:	fasted at time of dosing	fasted at time of dosing	fasted at time of dosing	fasted at time of dosing
Test item:	[^{14}C]linagliptin	[^{14}C]linagliptin	[^{14}C]linagliptin	[^{14}C]linagliptin
Vehicle/Formulation:	solution	solution	solution	solution
Method of Administration:	single intravenous bolus	single oral bolus	single intravenous bolus	single oral bolus
Dose:	2 mg/kg	2 mg/kg	2 mg/kg	2 mg/kg
Radionuclide:	^{14}C	^{14}C	^{14}C	^{14}C
Specific Activity [a]:	0.95 MBq/µmol	0.60 MBq/µmol	0.95 MBq/µmol	0.60 MBq/µmol
Sampling times:	3, 24, 168[b] h	168[b] h	3, 24, 168[b] h	168[b] h

[a] calculated from reported data [b] including excretion balance

Tissue/organ	**Radioactivity concentration [nmol/kg]**									
	3 h	**24 h**	**168 h**	**168 h**	**168 h**	**3 h**	**24 h**	**168 h**	**168 h**	**168 h**
brain (total)	BLD	BLD	BLD	BLD	BLD	BLD	BLD	BLD	BLD	BLD
salivary gland	1911	144	BLD	BLD	BLD	3009	537	73	69	77
brown fat	470	109	17*	BLD	BLD	930	105	23*	BLD	BLD
thymus	1139	38	BLD	BLD	BLD	1802	356	124	126	144
lungs	1073	33	BLD	BLD	BLD	1492	520	134	106	120
heart muscle	384	30	BLD	BLD	BLD	631	86	20*	16*	BLD
blood (heart)	120	16*	BLD	BLD	BLD	124	BLD	BLD	BLD	BLD
liver	3814	1306	136	74	112	5304	2029	499	379	417
spleen	1707	131	58	BLD	BLD	2983	603	145	91	106
pancreas	1518	94	BLD	BLD	BLD	1876	185	52	32	35
inner medulla	2193	147	24*	BLD	BLD	2787	296	234	173	192
outer medulla	1852	153	63	BLD	27*	12116	9468	8614	7811	8419
kidney cortex	1558	232	127	20*	35	7047	4849	2662	2299	2516
kidney total	1755	201	82	BLD	22*	6898	4732	3199	3133	3325

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5.I Pharmacokinetics: Organ Distribution (Continued)

Report No.: **U07-1488**

Species/Strain:	rat / F344 (DPP-4 deficient)	rat / F344 (DPP-4 deficient)	rat / F344 (wildtype)	rat / F344 (wildtype)
Method of Administration:	single intravenous bolus	single oral bolus	single intravenous bolus	single oral bolus

Tissue/organ	**Radioactivity concentration [nmol/kg]**									
	3 h	**24 h**	**168 h**	**168 h**	**168 h**	**3 h**	**24 h**	**168 h**	**168 h**	**168 h**
adrenal total	1074	79	24*	BLD	BLD	1426	212	67	42	50
fat	28*	BLD	BLD	BLD	BLD	78	17*	BLD	BLD	BLD
testis	298	202	19*	BLD	BLD	350	217	BLD	BLD	BLD
epididymis	878	163	BLD	BLD	BLD	1482	473	194	136	143
skin (total)	535	44	BLD	BLD	BLD	735	121	33	29*	30*
bone marrow	717	49	BLD	BLD	BLD	883	115	30*	BLD	17*
tongue	425	27*	BLD	BLD	BLD	570	31	BLD	BLD	BLD
muscle (caudal)	619	18*	BLD	BLD	BLD	912	22*	BLD	BLD	BLD
urinary excretion [% of dose]	ND	ND	23.1%	4.7%	8.9%	ND	ND	20.2%	6.6%	5.6%
faecal excretion [% of dose]	ND	ND	71.8%	87.6%	82.7%	ND	ND	95.4%	89.5%	86.7%

BLD = below limit of detection *between limit of quantitation and limit of detection ND = not determined

Additional information:
The distribution of radioactivity in male DPP-4-deficient and non-deficient Fischer rats was studied after intravenous or oral administration of 2 mg/kg [^{14}C]linagliptin. The comparison of the distribution pattern of radioactivity in non-deficient and DPP-4-deficient rat indicates that [^{14}C]linagliptin and/or its metabolites bind specifically and strongly to DPP-4 in the outer medulla of the kidney after dosing of 2 mg/kg [^{14}C]linagliptin. Other DPP-4-specific binding sites for drug-related radioactivity after dosing of 2 mg/kg [^{14}C]linagliptin to rats seemed to be at least in liver, lung, thymus, spleen and epididymis. Data of this study have been published [P09-08326].

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5.J PHARMACOKINETICS: ORGAN DISTRIBUTION

Test Article: Linagliptin

Location in Module 4: 4.2.2.3
Report No.: U07-1985
Study No.: A135/07GR

Species/Strain:	rat / Wistar (Crl:WI Han)	rat / Wistar (Crl:WI Han)	rat / Wistar (Crl:WI Han)
Gender (M/F)/Number of animals:	M / 4	M / 3	M / 3
Feeding condition:	fasted at time of dosing	fasted at time of dosing	fasted at time of dosing
Test item:	[^{3}H]linagliptin	[^{3}H]linagliptin	[^{3}H]linagliptin
Vehicle/Formulation:	solution	solution	solution
Method of Administration:	single intravenous bolus	single intravenous bolus	single intravenous bolus
Dose:	0.0074 mg/kg	0.1 mg/kg	2 mg/kg
Radionuclide:	^{3}H	^{3}H	^{3}H
Specific Activity [a]:	3412 MBq/µmol	256 MBq/µmol	13.4 MBq/µmol
Sampling times:	0.033, 3, 24[b], 168 h	3, 24[b], 168 h	3, 24[b], 168 h

[a] calculated from raw data [b] including urinary excretion sampling

Tissue/organ	**Radioactivity concentration [nmol/kg]**									
	0.033 h	**3 h**	**24 h**	**168 h**	**3 h**	**24 h**	**168 h**	**3 h**	**24 h**	**168 h**
brain (total)	0.0	0.0	0.0	0.0	0	0	0	0	0	0
salivary gland	13.9	23.2	16.3	7.4	137	62	20	1173	107*	0
brown fat	5.1	0.9	0.3*	0.0	11	5*	0	832	0	0
thymus	2.9	3.6	6.2	7.5	103	76	69	651	271	122*
lungs	199.1	132.3	60.7	14.5	410	194	68	1350	293	91*
heart muscle	47.1	28.7	9.7	1.6	81	28	8	305	0	0
liver	26.9	45.2	62.2	40.6	493	313	191	2441	790	219
spleen	21.9	47.3	58.0	19.4	505	228	117	1552	329	116*
pancreas	22.9	15.0	6.2	2.2	70	43	14	1063	91*	0
inner medulla	22.1	15.8	17.9	34.1	160	80	76	1073	159	112*
outer medulla	26.5	17.7	36.2	46.9	2678	3193	3803	5696	3835	3394
kidney cortex	127.7	100.5	108.7	90.5	1153	1029	717	2064	1186	714
kidney total	70.1	56.8	61.7	70.3	1126	1244	1618	2755	1513	1210

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5.J Pharmacokinetics: Organ Distribution (Continued)

Report No.: U07-1985

Dose:	0.0074 mg/kg				0.1 mg/kg			2 mg/kg		
Tissue/organ	**Radioactivity concentration [nmol/kg]**									
	0.033 h	**3 h**	**24 h**	**168 h**	**3 h**	**24 h**	**168 h**	**3 h**	**24 h**	**168 h**
adrenal total	65.4	49.9	25.3	5.2	153	105	25	627	209	0
fat	0.0	0.0	0.0	0.0	0	0	0	0	0	0
testis	0.3*	0.4*	0.4*	0.0	4*	0	0	186	89*	0
epididymis	3.3	5.2	6.2	2.6	69	NOS	20	NOS	281	0
skin (total)	2.6	3.4	4.7	2.8	88	67	17	436	91*	0
bone marrow	15.0	20.5	16.6	4.2	166	67	27	816	119*	0
tongue	8.4	3.6	1.5	0.3*	19	4*	0	239	0	0
muscle (caudal)	6.5	3.6	1.0	0.2*	18	4*	0	418	0	0
urinary excretion [% of dose]	ND	ND	0.93%	ND	ND	2.33%	ND	ND	20.32%	ND

* between limit of quantitation and limit of detection
NOS = no sample
ND = not determined

Additional information:

The distribution of radioactivity in male albino rats was studied after intravenous administration of three different doses of [^{3}H]linagliptin. At the low dose of 0.0074 mg/kg the distribution pattern of the radioactivity was distinctly different to that at higher doses (0.1 and 2 mg/kg). The tissue distribution pattern at the lowest dose fits impressively well with the tissue distribution pattern of DPP-4 and thus reflects probably the strong and long lasting binding of [^{3}H]linagliptin to DPP-4. Specific binding sites for drug-related radioactivity after intravenous dosing of [^{3}H]linagliptin to rats seemed to be at least in kidney, liver, lungs, spleen, and thymus. Already at a dose of about 0.1 mg/kg these binding sites seemed to be at least partly saturated in some tissues especially in kidney.

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5.K PHARMACOKINETICS: ORGAN DISTRIBUTION

Test Article: Linagliptin

Location in Module 4: 4.2.2.3
Report No.: U08-1913
Study No.: A136/07GR

Species/Strain:	rat	rat	rat	rat
Strain	Wistar Crl:WI(Han)	Wistar Crl:WI(Han)	Wistar Crl:WI(Han)	Wistar Crl:WI(Han)
Gender (M/F)/Number of animals:	M / 3	M / 3	M / 3	M / 1
Feeding condition:	fasted at time of dosing	fasted at time of dosing	fasted at time of dosing	fasted
Test item:	[^{3}H]linagliptin	[^{3}H]linagliptin	[^{3}H]linagliptin	-
Vehicle/Formulation:	solution	solution	solution	-
Method of Administration:	single intravenous bolus	single intravenous bolus	single intravenous bolus	-
Dose:	0.0074 mg/kg	0.1 mg/kg	2 mg/kg	-
Radionuclide:	^{3}H	^{3}H	^{3}H	-
Specific Activity [a]:	3083 MBq/µmol	236 MBq/µmol	12.3 MBq/µmol	-
Sampling times:	0.033, 3, 192 h	3 h	3 h	-

[a] calculated from raw data

Additional information:
Using micro autoradiography, qualitative distribution of radioactivity in kidney, liver and small intestine was investigated in dependence on time after intravenous bolus administration of 0.0074 mg/kg [^{3}H]linagliptin (2 min, 3 h, 192 h). In addition, the dose dependence of qualitative distribution of radioactivity in kidney was investigated by intravenous bolus administration of 0.0074, 0.1 and 2 mg/kg [^{3}H]linagliptin.

The qualitative distribution pattern of the radioactivity in rat kidney changed with time after intravenous bolus administration of 0.0074 mg/kg [^{3}H]linagliptin, as the maximum concentration of radioactivity shifted from glomeruli (2 min) to the upper part (3 h) and lower part (192 h) of the proximal tubules. The radioactivity in the proximal tubules was primarily located in the brush border. In addition, it was shown that the distribution pattern in the kidney at 3 h after intravenous bolus administration of [^{3}H]linagliptin was dose-dependent. In the liver, a retiform distribution of the radioactivity was observed over the whole time interval investigated after intravenous bolus administration of 0.0074 mg/kg [^{3}H]linagliptin with prominent radioactivity concentration in the acinus regions around the portal triads. In the small intestine, a shift of the radioactivity with time from the lamina propria (dominant at 2 min) to the surface of the villi (dominant at 3 h and 192 h) was indicated after intravenous bolus administration of 0.0074 mg/kg [^{3}H]linagliptin.

In general, the distribution pattern of [^{3}H]linagliptin related radioactivity on a cellular level closely resembled the known distribution of DDP-4. This observation together with the persistency of binding indicate that linagliptin binds specifically, reversibly and with high affinity to its target DPP-4 in tissues.

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5.L PHARMACOKINETICS: ORGAN DISTRIBUTION

Test Article: Linagliptin

Location in Module 4: 4.2.2.3
Report No.: U08-2101
Study No.: BOI/0331

Species: **Cynomolgus monkey (from 52 week toxicity study no. BOI/0331)**
Gender (M/F) / Number of animals: **M/20, F/20**
Feeding condition: **ad libitum**
Vehicle/Formulation: **solution in tap water**
Method of Administration: **repeated oral (gavage) administration for 52 weeks**
Dose levels (mg/kg): **0, 1, 10, 100**
Sampling time: **at necropsy of 52 week toxicity study**

Group	Dose [mg/kg]	mean kidney linagliptin [nmol/kg]	CV (%)	mean kidney CD 1790 [nmol/kg]	CV (%)	mean liver linagliptin [nmol/kg]	CV (%)	mean liver CD 1790 [nmol/kg]	CV (%)
1	0	1160	31.2	BLQ	NC	BLQ	NC	BLQ	NC
1 (recovery)	0	8510	56.6	BLQ	NC	1030	NC	BLQ	NC
2	1	154000	10.8	BLQ	NC	7710	19.2	4780	52.6
3	10	229000	22.2	BLQ	NC	52500	52.2	45600	59.2
4	100	1590000	63.6	9110	56.9	1670000	71.1	123000	32.3
4 (recovery)	100	133000	7.0	BLQ	NC	9450	25.0	BLQ	NC

BLQ = below limit of quantification
NC = not calculated

In conclusion, linagliptin is distributed extensively into kidney and liver after oral administration to Cynomolgus monkeys. linagliptin concentrations in kidney were markedly higher than in liver at 1 and 10 mg/kg, but not at 100 mg/kg. In kidneys a pronounced non-linear increase of the linagliptin concentration with dose was observed, whereas liver concentrations increased virtually dose-proportionally. This can be explained by saturable binding of linagliptin to DPP-4 in the kidney which was previously demonstrated for mice and rats. In contrast, dose-proportional distribution of linagliptin into liver predominated the tissue distribution in this organ. Additional exposure to CD 1790 was demonstrated in the liver, whereas in the kidney the CD 1790 concentrations were quantifiable only at the highest dose of 100 mg/kg. linagliptin in kidney and liver decreases markedly after end of dosing, however, a long terminal half-life of linagliptin due to specific binding to its target DPP-4 in kidney and tissue is suggested.

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5.M PHARMACOKINETICS: ORGAN DISTRIBUTION

Test Article: Linagliptin

Species:	rabbit Crl:CHBB(HM)	Cynomolgus monkey
Gender (M/F) / Number of animals:	F / 3	M / 2 & F / 2
Feeding condition:	ad libitum	ad libitum
Vehicle/Formulation:	acidified solution in tap water	solution in tap water
Method of Administration:	oral (gavage)	oral (gavage) / intraduodenal under anaesthesia
Dose levels (mg/kg):	25	5
specific radioactivity	about 0.051 MBq/µmol	about 0.24 MBq/µmol
Sampling time:	at end of plasma or terminal bile sampling in ADME study	about 6h after i.d. dosing for terminal bile sampling in ADME study (1M/1F) or about 168 h after oral dosing for mass balance
Study design:	autoradiography of explanted kidneys from 3 rabbits, which were previously included into a mass balance and plasma/blood level experiment. Evaluation only qualitatively.	autoradiography of explanted kidneys from 2 male and 2 female cynomolgus monkeys, which were previously included into a mass balance and biliary excretion experiment. Evaluation only qualitatively.
Location in Module 4:	4.2.2.5	4.2.2.2
Report-No.:	U06-1087	U06-1132
Study No.:	A032/05FU	AA27415
Result:	Like in rats, the radioactivity concentration was higher in the renal cortex than in the medulla of female rabbits and showed also a long retention time as suggested by the still noticeable radioactivity found at 168 h after administration. However, a distinctly labelled intermediate zone was only observed at 1 hour but apparently absent at 4 and 168 hours. Thus, in contrast to rats, the tripartite distribution pattern was not observed in rabbits.	The outer medulla/inner cortex region showed the same radioactivity concentrations as the cortex, indicating no distinctly labelled intermediate zone between the renal cortex and the medulla, which was observed in rats.

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6.A PHARMACOKINETICS: PLASMA PROTEIN BINDING

Test Article: Linagliptin

Location in Module 5: **5.3.2.1**
Report No.: **U04-1420**
Study No.: **A016/04FU**

Study system: plasma
Target entity, Test system and method: [^{14}C]linagliptin, equilibrium dialysis with LSC

Species	Conc. tested	% Bound
rat [Crl:WI(Han)]	21.1 - 3180 nM	87.4 - 80.7
dog [Beagle]	21.7 - 3020 nM	89.7 - 86.4
minipig [Göttingen]	206 nM	58.7
human	30.7 - 5230 nM	89.3 - 84.0

Additional Information:

The plasma protein bound fractions of [^{14}C]linagliptin related radioactivity (mean values overall concentrations and genders) were between 58.7% and 87.8% [minipigs (58.7%) << rats (83.9%) < humans (84.7%) < dogs (87.8%)]. Hence, a pronounced species difference was observed between minipigs and the other investigated species. Minor but consistent effects of drug concentration on the fraction of protein bound drug were observed in rats, dogs and humans, where the fraction of protein bound linagliptin related radioactivity decreased slightly (5.3, 3.3 or 6.7% difference between the highest and the lowest [^{14}C]linagliptin concentration tested, human, dog and rat, respectively) with increasing concentrations of [^{14}C]linagliptin at concentrations between 20 and 3000 nM (5000nM in humans). This may be addressed to a saturation of binding sites.

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6.B PHARMACOKINETICS: PLASMA PROTEIN BINDING

Test Article: Linagliptin

Location in Module 4: 4.2.2.3
Report No.: U05-2596
Study No.: A033/05FU

Study system: plasma
Target entity, Test system and method: [^{14}C]linagliptin, equilibrium dialysis with LSC

Species / Strain	Conc. tested	% Bound
Mouse /CD-1 [Crl:CD1(ICR)]	30 - 3000 nM	78.2 - 72.7
Rabbit / Himalayan [Crl:Chbb(HM)] (females only)	30 - 3000 nM	84.3-79.6
Cynomolgus monkey	30 - 3000 nM	82.0 - 70.4
Human	1 - 30 nM	96.3 - 88.5

Additional Information:
[^{14}C]linagliptin shows a moderate binding to plasma proteins in the investigated species at concentrations of 300 nmol/L and higher. The mean plasma protein binding in this concentration range was between about 70 and 80% in mouse, rabbit and cynomolgus monkeys. A considerable concentration dependency was observed, with increasing fractions of protein bound linagliptin with decreasing plasma concentrations (between 3000 and 30 nmol/L). In human plasma, maximum protein binding was high (about. 97%) and observed at concentration of 3 nmol/L and lower.

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6.C PHARMACOKINETICS: PLASMA PROTEIN BINDING

Test Article: Linagliptin

Location in Module 4: 4.2.2.3
Report No.: U06-2155
Study No.: A061/06FU

Study system: plasma
Target entity, Test system and method: [^{14}C]linagliptin, equilibrium dialysis with LSC

Species / Strain	Conc. tested	% Bound
Rat / Wistar [Crl:WI(Han)]	3, 10, 30 nM	96.1, 83.5, 76.8
Rat / wildtype Fischer [F344/DuCrl]	3, 10, 30 nM	95.4, 85.6, 80.7
Rat / DPP-4 deficient Fischer [F344/DuCrlCrlJ]	3, 10, 30 nM	70.7, 71.7, 70.9

Additional Information:

Plasma protein binding of [^{14}C]linagliptin in rat plasma was different between 3 and 30 nM: the bound fraction was lower at 30 nM (about 80%), the binding markedly increased to about 96% at 3 nM. No relevant differences between wild-type-like Fischer and Wistar rats were observed. In contrast, the concentration dependency was totally absent in plasma of the DPP-4 deficient Fischer rats. The bound fraction was about 70% between 3 and 30 nM. In addition, the bound fraction was markedly lower at 30 nM in DPP-4 deficient Fischer rat plasma as compared to wild-type like Fischer rat plasma.
These data show, that saturable binding to plasma DPP-4 is the reason for the concentration dependency in linagliptin plasma protein binding in rats.

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6.D PHARMACOKINETICS: PLASMA PROTEIN BINDING

Test Article: Linagliptin

Location in Module 5: 5.3.2.1
Report No.: U08-1216
Study No.: A083/07FU

Study system: plasma
Target entity, Test system and method: [^{3}H]linagliptin, equilibrium dialysis with LSC

Species / Strain	Conc. tested	% Bound
Mouse / wildtype [C57BL/6J]	0.172 - 13100 nM	max: 99.3 ± 0.6 min: 74.5 ± 0.3 (nonlinear regression)
Mouse / DPP-4 knockout [C57BL/6-TgH(CD26)-CIML]	0.0628 - 14300 nM	~70 - 80 no concentration dependency
Rat / wildtype Fischer [F344/DuCrl]	0.139 - 12700 nM	max: 99.2 ± 0.8 min: 77.5 ± 0.5 (nonlinear regression)
Rat / DPP-4 deficient Fischer [F344/DuCrlCrlJ]	0.0636 - 13800 nM	~60 - 80 no concentration dependency
Human	0.021 - 29900 nM	max: 99.3 ± 0.2 min: 77.3 ± 0.3 (nonlinear regression)
human alpha-1-acid-glycoprotein in buffer (1g/L)	2279, 34646 nM	32.8, 2.6
human serum albumin in buffer (45 g/L)	644, 18395 nM	48.2, 37.0

Additional Information:
The present study demonstrated that the plasma protein binding of the DPP-4 inhibitor linagliptin in human plasma and plasma of wildtype mice and rats shows a prominent concentration dependency which is lacking in DPP-4 deficient mice and rats. Thus, the concentration dependency is shown to originate from saturation of the binding of linagliptin to soluble DPP-4 in plasma. The maximum fraction bound of linagliptin is around 99% in mouse, rat and human plasma. The binding of linagliptin to DPP-4 is of very high affinity with an affinity constant of about 2 - 3 $\bullet 10^{10}\,M^{-1}$ in wildtype mouse, rat and human plasma. Low binding of linagliptin to human albumin (f_B = 48.2%) and human alpha-1 acid glycoprotein (f_B = 32.8%) was observed in vitro using isolated purified proteins. Parts of this study have been published [P09-00166].

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6.E PHARMACOKINETICS: PLASMA PROTEIN BINDING

Test Article: Linagliptin

Location in Module 5: **5.3.2.1**
Report No.: **U09-1951**
Study No.: **A105/09FU**

Study system: plasma
Target entity, Test system and method: CD 1790, equilibrium dialysis, sample analysis by HPLC-MS/MS

Species / Strain	Conc. tested	% Bound
Rat / Wistar [Crl:WI(Han)]	10,100 nM	89.3
Cynomolgus monkey	10,100 nM	97.1
human	10,100 nM	94.7

Additional Information:
A moderate to high plasma protein binding of CD 1790 was observed in rats, Cynomolgus monkeys and humans. Minor, but with regard to the unbound fraction relevant species differences were observed, with rats showing a lower protein binding of CD 1790 than Cynomolgus monkeys. The protein binding of CD 1790 in human plasma was in between these species. In the investigated concentration range, the plasma protein binding was constant, indicating non-saturable binding. At 1 nM CD 1790, the dialysate concentration of CD 1790 was BLQ.

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6.F PHARMACOKINETICS: PLASMA PROTEIN BINDING (DISTRIBUTION IN BLOOD)

Test Article: Linagliptin

Study system: blood
Target entity, Test system and method: [^{14}C]linagliptin,
method: measurement of hematocrit and radioactivity in blood and plasma
The concentration ratio of radioactivity in blood cells and plasma (C_C/C_P) was calculated.

Species	mouse (m&f)	rat (m)	rabbit (f)	rat (m&f)	dog (m&f)	human (m&f)
Report No.	U05-2607	U04-1731	U06-1087	U04-1420	U04-1420	U04-1420
Study No.	A034/05FU	A017/04FU	A032/05FU	A016/04FU	A016/04FU	A016/04FU
Location in Module 4 resp. 5:	4.2.2.5	4.2.2.5	4.2.2.5	5.3.2.1	5.3.2.1	5.3.2.1
conc. range [nmol/L]	286 – 28078	3.72 - 61.9	29.3 - 9220	233	264	299
study design	ex vivo after oral administration of 25 mg/kg [^{14}C]linagliptin	ex vivo after oral administration of 1 mg/kg [^{14}C]linagliptin	ex vivo after oral administration of 25 mg/kg [^{14}C]linagliptin	in vitro (spiked blood) 0 - 3 h after spiking	in vitro (spiked blood) 0 - 3 h after spiking	in vitro (spiked blood) 0 - 3 h after spiking
C_C/C_P	0.982 (overall mean)	1.281 (30 min) 0.303-0.666 (2 h – 24 h)	0.511 - 1.577	1.00 - 1.18	0.50 - 0.58	0.50 - 0.53
additional information	no time (concentration) dependency	C_C/C_P exceeding 1 only at 30 min, thereafter no prominent time dependency	higher values early after dosing	no time dependency	no time dependency	no time dependency

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6.G PHARMACOKINETICS: PLASMA PROTEIN BINDING (DISTRIBUTION IN BLOOD)

Test Article: Linagliptin

Location in Module 4: **5.3.2.1**
Report No.: **U09-2476**
Study No.: **A099/09FU**

Study system: fresh human blood, in vitro
Target entity, Test system and method: [^{14}C]linagliptin, spiked into blood
method 1: measurement of hematocrit and radioactivity in blood and plasma
method 2: measurement radioactivity in plasma and blood cells after centrifugation
The concentration ratio of radioactivity in blood cells and plasma (C_C/C_P) was calculated.

The geometric mean values of C_C/C_P are given in the Summary Table.

Target conc. of [^{14}C]linagliptin in blood:		1 nM	5 nM	10 nM	50 nM	100 nM	500 nM	1000 nM
method 1	**gMean**	0.136	0.144	0.256	0.415	0.429	0.484	0.490
	gCV (%)	88.8	43.4	15.7	8.3	4.9	2.5	4.0
method 2	**gMean**	0.119	0.192	0.295	0.465	0.459	0.532	0.522
	gCV (%)	37.8	12.0	6.4	6.0	5.7	6.4	4.6

It can be concluded, that at therapeutic plasma levels in humans, the distribution of linagliptin in blood depends on its concentration. Whereas at very low concentrations, linagliptin is almost completely distributed into the plasma space, at higher concentrations a slight distribution into or onto blood cells occurs. The concentration dependency is most likely due to changes in plasma protein binding. The change in plasma protein binding was demonstrated to be due to specific, high affinity but low capacity, i.e. saturable binding to its target DPP-4, which is present in plasma in a soluble form.

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6.H PHARMACOKINETICS: PLASMA PROTEIN BINDING (DISTRIBUTION IN BLOOD)

Test Article: Linagliptin

Location in Module 4: **4.2.2.3**
Report No.: **U10-1144**
Study No.: **A110/09FU**

Study system: fresh rat and cynomolgus monkey blood, in vitro
Target entity, Test system and method: [^{14}C]linagliptin, spiked into blood
method 1: measurement of hematocrit and radioactivity in blood and plasma
method 2: measurement radioactivity in plasma and blood cells after centrifugation
The concentration ratio of radioactivity in blood cells and plasma (C_C/C_P) was calculated.

The geometric mean values of C_C/C_P are given in the Summary Table.

		Rat		Cynomolgus monkey	
Target conc. blood:		1 nM	300 nM	1 nM	300 nM
method 1	mean	NC	1.06	NC	0.582
method 2	mean	0.0856	1.03	0.0544	0.546

At 300 nM linagliptin shows an almost equal distributed between blood cells and plasma in rat blood. In addition, a relevant distribution into blood cells was observed in Cynomolgus monkey blood at this concentration. In contrast, the compound was virtually exclusively restricted to the plasma space at 1 nM in both species in vitro.
In conclusion, the observed concentration dependency in the distribution of linagliptin in rat and cynomolgus monkey blood is consistent with its concentration dependent plasma protein binding. This is due to high affinity but low capacity binding of linagliptin to plasma DPP-4.

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7.A PHARMACOKINETICS: STUDY IN PREGNANT OR NURSING ANIMALS

Test Article: Linagliptin

Location in Module 4: 4.2.2.3
Report No.: U09-2235
Study No. A179/08RB

Placental transfer

Species:	Rat (Wistar)	
Strain	Wistar	
Number of animals:	F /5	F / 5
Gestation day /	11	19
Vehicle/Formulation:	aqueous solution acidified with HCl	
Method of Administration:	oral (gavage)	
Dose (mg/kg):	1 mg/kg	
Analyte:	Radioactivity	
Assay:	Whole body autoradiography	

Mean concentration of radioactivity:		[nmol/kg tissue]					[nmol/kg tissue]				
	Sampling time:	**1 h**	**2 h**	**8 h**	**24 h**	**48 h**	**1 h**	**2 h**	**8 h**	**24 h**	**48 h**
Organs and tissues of **dams**	Harder's gland	14.4*	20	71.1	16.1	12.8*	0	23.5	86.2	24.1	9.44*
	tongue	14.2*	21.7	29.4	0	9.6*	0	23.7	30.1	16	11.3*
	brain	0	0	0	0	0	0	0	0	0	0
	pituitary	17.9	26.5	46.2	11.0*	8.12*	0	25.3	58.1	15.5*	11.2*
	brown fat	23.8	19.1	34.3	11.7*	10*	0	15.9	22.1	11.0*	0
	salivary gland	36.6	69.2	208	126	84.9	0	68.2	157	181	114
	thymus	10.6*	16	117	108	172	0	19.7	174	290	239
	heart muscle	71.9	75.9	76	40.4	35.9	13.0*	86.1	95.8	65	44.8
	lungs	350	519	565	301	295	134	529	685	611	350
	liver	899	948	1000	639	690	389	892	955	701	647
	spleen	162	253	521	289	239	23.4	226	614	376	221
	adrenal	151	251	385	213	143	12.2*	289	257	229	190
	kidney	303	420	2880	2660	4150	26.2	394	3090	5490	4970
	pancreas	39.2	44.6	99.8	54.3	62.8	0	49.4	91.2	80.1	60.3
	bone marrow	23.2	37.8	104	38.2	38.5	0	34.4	101	69.7	41.3
	muscle	0	0	10.8*	0	0	0	0	16.1	0	0
	fat	0	0	0	19.1	37.6	0	0	26.6	15.6	8.04*
	skin (back)	0	0	54.2	50.3	49.3	11.4*	0	41	41.7	28.4
	blood	9.34	8.76	4.99	2.71	1.31	3.72	5.98	4.69	1.67	0.915
	plasma	13.5	12.2	6.96	3.50	2.44	4.88	6.90	3.31	2	0.869

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7.A Pharmacokinetics: Organ Distribution (Continued)

Report No.: U09-2235

Mean concentration of radioactivity:		[nmol/kg tissue]					[nmol/kg tissue]				
		1 h	**2 h**	**8 h**	**24 h**	**48 h**	**1 h**	**2 h**	**8 h**	**24 h**	**48 h**
Placenta and organs and tissues of **fetus**	placenta	23.9	NOS	14.2*	0	13.3*	0	9.61*	13.4*	0	0
	embryo	17.1	NOS	12.4*	0	0	NA	NA	NA	NA	NA
	fetus	NA	NA	NA	NA	NA	0	0	0	8.97*	0
	fetus brain	NA	NA	NA	NA	NA	ND	ND	0	0	0
	fetus spinal cord	NA	NA	NA	NA	NA	ND	ND	ND	0	0
	fetus heart	NA	NA	NA	NA	NA	ND	0	0	9.09*	0
	fetus lungs	NA	NA	NA	NA	NA	ND	ND	ND	0	0
	fetus liver	NA	NA	NA	NA	NA	0	0	13.5*	22.5	13.6*

*: uncertain data due to low concentration LOQ: 16 nmol/kg LOD: 8 nmol/kg (values below LOD were set to 0)

NA = not applicable

ND = not determined as respective concentrations could not be evaluated

NOS = no sample

Additional Information:

Gestation day 11: In the embryos and placentas, concentrations from below the lower limit of detection (8 nmol/kg) to about the lower limit of quantification (16 nmol/kg) were found. In many tissues of the dams an elimination phase became evident from 8 to 48 h, in kidney and thymus, however, no decrease was seen up to 48 h

Gestation day 19: In the fetuses similar concentrations were seen than in the embryos at the respective time-points on gestation day 11, traces of drug related radioactivity could be detected in fetal heart and fetal liver.

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7.B PHARMACOKINETICS: STUDY IN PREGNANT OR NURSING ANIMALS

Test Article: Linagliptin

Location in Module 4: 4.2.2.3
Report No.: U10-1332
Study No. 09B138

Species: Rat
Strain Crl:WI(Han)
Gestation day 16
Number of animals: F/ 27 (30 mg/kg) or F / 28 (240 mg/kg)
Vehicle/Formulation: suspension with Natrosol
Method of Administration: oral (gavage)
Dose (mg/kg): 30 and 240 mg/kg
Analyte: linagliptin
Assay: HPLC-MS/MS

Study design: The test item was administered to pregnant rats once daily as suspension by oral gavage from gestation Day 7 (= Dosing Day 1) to gestation Day 16 (= Dosing Day 10). The systemic exposure was assessed by measuring plasma concentrations of linagliptin on dosing Day 1 in the dams (data not shown) and at 1, 3, 8 and 24 h on Dosing Day 10 in dams and corresponding embryos after hysterectomy.

		30 mg/kg			240 mg/kg		
Parameter	**Unit**	maternal	embryonal	ratio embryonal/maternal (%)	maternal	embryonal	ratio embryonal/maternal (%)
C_{max}	**[nM]**	1960	526	**26.8**	13000	5050	**38.8**
AUC_{0-24h}	**[nM·h]**	10600	4560	**43.0**	146000	79000	**54.1**

In conclusion, linagliptin crosses the blood placenta barrier in rats. Substantial exposure to linagliptin in embryos on gestation Day 16 was achieved after once daily oral administration of 30 or 240 mg/kg linagliptin to pregnant rats. The embryonal exposure reached about 50% of the maternal exposure, whereas embryonal C_{max} was up to about 40% of the maternal C_{max}. It can therefore be concluded, that relevant exposure to linagliptin of the embryo was also achieved in the respective embryo-fetal development studies at these dose levels.

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7.C PHARMACOKINETICS: STUDY IN PREGNANT OR NURSING ANIMALS

Test Article: Linagliptin

Location in Module 4: 4.2.2.3
Report No.: U10-1684
Study No. C73207

Species:	Rabbit
Strain	Himalayan Crl:CHBB(HM)
Gestation day	18
Number of animals:	F/ 22 per dose group
Vehicle/Formulation:	suspension with Natrosol
Method of Administration:	oral (gavage)
Dose (mg/kg):	25 and 150 mg/kg
Analyte:	linagliptin, CD 1790
Assay:	HPLC-MS/MS

Study design: The test item was administered once daily as suspension by gavage from gestation Day 6 (GD 6) through gestation Day 18 (GD 18). Linagliptin dose levels were 25 and 150 mg/kg/day. The systemic exposure of the does was assessed by measuring plasma concentrations of linagliptin and CD 1790 on GD 6 (= dosing Day 1) before treatment and at 1, 2, 3, 4, 8 and 24 h after administration. On GD 18 (= dosing Day 13), the animals were divided into 4 subgroups. Caesarian section of the does of each subgroup was performed at 1, 3, 8 or 24 h after dosing and maternal and fetal blood samples were taken at the respective time points. Prior to this, blood samples were taken from the does in-life using the same sampling schedule as on GD 6.

			25 mg/kg linagliptin			**150 mg/kg linagliptin**		
	Parameter	**Unit**	maternal (GD18)	fetal (GD18)	ratio embryonal/maternal (%)	maternal (GD18)	fetal (GD18)	ratio embryonal/maternal (%)
linagliptin	**C_{max}**	**[nM]**	5680	96.5	**1.7**	33200	1690	**5.1**
	AUC_{0-24h}	**[nM·h]**	20600	518	**2.5**	371000	16300	**4.4**
CD 1790	**C_{max}**	**[nM]**	244	41.6	**17.0**	1500	412	**27.5**
	AUC_{0-24h}	**[nM·h]**	787	119	**15.1**	13900	3290	**23.7**

It is concluded, that the data obtained in the present supportive study are suitable to estimate exposure to CD 1790 in pregnant rabbits (does) and exposure to linagliptin and CD 1790 in the fetus on gestation Day 18 of the pivotal embryo-fetal development study in rabbits. However, due to the slightly lower exposure achieved in the 25 mg/kg group of the pivotal study, the maternal to fetal ratios of C_{max} and AUC_{0-24h} for linagliptin and CD 1790 determined in the present study at 25 mg/kg should be used for retrospective estimation of C_{max} and AUC_{0-24h} of both analytes in the pivotal study.

GLP compliance: yes

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7.D PHARMACOKINETICS: STUDY IN PREGNANT OR NURSING ANIMALS

Test Article: Linagliptin

Location in Module 4: 4.2.2.5
Report No.: U08-1929
Study No. A174/07RB

Species:	Rat		
Strain:	Crl:WI(Han)		
Lactating date	day 14		
Number of animals:	F/ 5		
Feeding condition:	fed		
Vehicle/Formulation:	aqueous solution acidified with HCl, ^{14}C-labelled		
Method of Administration:	oral (gavage)		
Dose:	30 mg/kg		
Analyte:	radioactivity, linagliptin and CD 1790, metabolites		
Assay:	LSC for radioactivity, LC-MS/MS for linagliptin and CD 1790, LC-MS (QoaTOF) for metabolites		
Time after dosing [h]	**1 h**	**6 h**	**24 h**
Concentration:	[nmol/L]	[nmol/L]	[nmol/L]
Milk:	6830	1390	31.7
Plasma:	1620	285	37.4
Milk / plasma ratio:	4.22	4.88	0.85
Neonates:	not detected	not detected	not detected

Additional Information:
The amount of parent compound excreted via milk within 24 h after drug administration was estimated to 0.32% of the dose. The remaining radioactivity excreted via milk (0.028% of the dose) was distributed over 3 - 7 minor metabolites. Individually they accounted for ≤ 0.008% of the dose within 24 h.

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8. PHARMACOKINETICS: OTHER DISTRIBUTION STUDY PERMEABILITY IN CACO-2 CELLS

Test Article: Linagliptin

Location in Module 4: **4.2.2.2**
Report No.: **U05-1795**
Study No.: **C-10165-040-0404, B2566**

Type of study: **Passive absorption and active (P-glycoprotein (P-gp) mediated) transport, inhibition of P-gp**
Test item: **Linagliptin**

Method: The permeability characteristics of linagliptin was investigated by using the human colon carcinoma derived cell line CaCo-2. The CaCo-2 cell line constitutively expresses the efflux transporter P-gp. Therefore, the active transport of linagliptin by P-gp and the inhibition of the P-gp-mediated digoxin transport by linagliptin was investigated using CaCo-2 cells. In vitro apical-to-basal (a-to-b) and basal-to-apical (b-to-a) permeability studies were performed across CaCo-2 cell monolayers.

Results: The test system was validated by using reference compounds mannitol, atenolol and propranolol as marker substrates for low, medium and high passive permeability. The functional expression of P-gp was assessed by using the P-gp substrates digoxin and WEB 2086 BS. All test substrates yielded acceptable passive permeability data or active P-gp-mediated transport. linagliptin showed apically-directed vectorial transport and concentration-dependent permeability that indicated the presence of active transport. Inhibitors of P-gp abolished the asymmetric permeability of linagliptin. linagliptin inhibited the P-gp mediated transport of digoxin in a concentration-dependent manner. A mean apparent IC_{50} was estimated of 55 µM.

Tabulated results:

Permeability in CaCo-2 cells and effects of P-gp inhibitors (arithmetic mean data (N = 3), CV(%) in brackets)

compound	apparent permeability [nm/sec] a to b direction	apparent permeability [nm/sec] b to a direction	efflux ratio
mannitol	5.38 (8.2)	6.32 (25.2)	1.17 (26.4)
atenolol	12.1	17.0	1.4
propranolol	201	188	0.94
WEB 2086 BS	4.61 (11.8)	105 (7.0)	22.96 (4.8)
digoxin	3.04 (5.7)	66.4 (14.0)	21.71 (8.4)
linagliptin, 3 µM*	1.80 (5.3)	122 (2.6)	67.82
linagliptin, 300 µM*	7.77 (8.3)	67.2 (14.2)	8.65
linagliptin + cylosporin A (12 µM)	27.5 (7.6)	41.1 (29.7)	1.47(22.0)
linagliptin + verapamil (200 µM)	27.3 (19.2)	46.5 (21.4)	1.71 (10.0)

* target concentration

Effect of linagliptin on the efflux ratio of digoxin

Concentration of linagliptin [µM]	digoxin efflux ratio
0	19.88
0.3	15.78
1	15.48
3	17.83
10	18.22
30	14.20
60	5.60
100	1.94
300	1.11

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9.A PHARMACOKINETICS: METABOLISM *IN VIVO* (METABOLITE PATTERN IN THE RAT)

Test Article: Linagliptin

Location in Module 4: **4.2.2.4**
Report No.: **U05-1981**
Study No.: **A234/04TE**

Species	Rat
strain	Wistar [Crl:WI(Han)] (former designation: CrlGlxBrlHan:WI)
Gender (M/F) / Number of animals	M / 5 & F/ 3: urine and faeces sampling after oral administration M / 5 & F/ 6: urine and faeces sampling after intravenous administration M / 3 & F/ 3: plasma and bile sampling (oral and intravenous administration) M / 4 & F/ 3: bile sampling after intravenous administration M / 3 & F/ 3: bile sampling after intraduodenal administration
Feeding condition:	fasted
Vehicle / Formulation:	solution in water, acidified with hydrochloric acid
Method of Administration:	oral via gavage / intraduodenal for bile sampling / intravenous bolus injection
Dose:	1.0 mg/kg
Radionuclide	^{14}C
Specific Activity	1.935 MBq/µmol (batch) about 0.65 MBq/µmol (dosed)

Additional information:
Mean data of male and female animals are presented since only minor gender related differences of the metabolite pattern in plasma, urine, faeces and bile were observed.

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9.A Pharmacokinetics: Metabolism *In Vivo* (Metabolite pattern in the rat, Continued)

Report No.: U05-1981

	oral administration of 1 mg/kg [^{14}C]linagliptin							
compound code	plasma, 1 h [nmol/l][a] (% of sample radioactivity)		plasma, 4 h [nmol/l] [b] (% of sample radioactivity)		urine 0 - 24 h	faeces 0 - 48 h	urine + faeces total	bile[b] 0 - 6 h
					[% of dose administered]			
Linagliptin (=BI 1356, M473(1), m0)	**136.4**	**(74.3)**	**91.6**	**(83.5)**	**1.1**	**69.9**	**71.0**	**24.8**
CD1790 (M474(1), (= m15)	11.5	(6.3)	13.9	(12.7)	0.1	0.7	0.7	-
M489(1) (= m9)	-	-	-	-	-	-	-	3.2
M489(2) (= m8)	-	-	-	-	-	0.6	0.6	1.6
M503(1), M665(1) (= m5)	-	-	-	-	-	-	-	3.4
M505(8), M505(9) (= m10)	-	-	-	-	-	1.6	1.6	3.1
M531(1), M531(2), M693(1) (= m12)	-	-	-	-	-	-	-	6.9
M707(1) (= m11)	-	-	-	-	-	-	-	1.3
m2	-	-	-	-	0.2	-	0.2	-
m3	-	-	-	-	-	-	-	2.8
m4	-	-	-	-	-	4.7	4.7	-
m6	-	-	-	-	-	2.4	2.4	-
m7	-	-	-	-	-	0.7	0.7	-
m14	35.5	(19.4)	8.4	(7.7)	-	2.3	2.3	-
m16	-	-	-	-	-	1.8	1.8	-
m17	-	-	-	-	0.1	-	0.1	-
sum	183.5	(100.0)	109.7	(100.0)	1.3	83.4	84.7	41.7
total excretion	-	-	-	-	1.3	94.1	95.3	-
not extracted / assigned	-	-	-	-	-	10.6	10.6	-

-: empty cells: not detected

a: plasma data of report U05-1981 was converted from [ng-eq/ml] into [nmol/l] using the molecular weight of [^{14}]linagliptin: 474.22 g/mol

b: after intraduodenal administration

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9.A Pharmacokinetics: Metabolism *In Vivo* (Metabolite pattern in the rat, Continued)

Report No.: U05-1981

	intravenous administration of 1 mg/kg [^{14}C]linagliptin							
compound code	plasma, 1 h [nmol/l] [a] (% of sample radioactivity)		plasma, 3 h [nmol/l] [a] (% of sample radioactivity)		urine 0 - 24 h	faeces 0 - 48 h	urine + faeces total	bile 0 - 6 h
					[% of dose administered]			
Linagliptin (=BI 1356, M473(1), m0)	**230.0**	**(90.9)**	**164.8**	**(90.6)**	**16.6**	**41.2**	**57.8**	**24.8**
CD1790 (M474(1), (= m15)	23.0	(9.1)	17.1	(9.4)	0.3	1.4	1.7	0.8
M489(1) (= m9)	-	-	-	-	-	1.3	1.3	2.0
M489(2) (= m8)	-	-	-	-	-	1.7	1.7	1.5
M503(1), M665(1) (= m5)	-	-	-	-	-	-	-	2.8
M505(8), M505(9) (= m10)	-	-	-	-	-	4.0	4.0	2.4
M531(1), M531(2), M693(1) (= m12)	-	-	-	-	-	-	-	7.2
M707(1) (= m11)	-	-	-	-	-	-	-	1.0
m1	-	-	-	-	0.2	-	0.2	-
m2	-	-	-	-	0.2	-	0.2	-
m3	-	-	-	-	-	0.7	0.7	1.4
m4	-	-	-	-	0.2	1.4	1.6	-
m6	-	-	-	-	-	1.0	1.0	-
m7	-	-	-	-	-	5.7	5.7	-
m13	-	-	-	-	0.4	-	0.4	-
m14	-	-	-	-	-	5.8	5.8	-
m16	-	-	-	-	-	2.9	2.9	-
m17	-	-	-	-	0.4	-	0.4	-
m18	-	-	-	-	0.1	-	0.1	-
sum	253.0	(100.0)	181.9	(100.0)	18.0	61.1	79.0	41.2
total excretion	-	-	-	-	18.0	71.0	89.0	-
not extracted / assigned	-	-	-	-	-	9.9	9.9	-

-: empty cells: not detected

[a]: plasma data of report U05-1981 was converted from [ng-eq/ml] into [nmol/l] using the molecular weight of [^{14}]linagliptin: 474.22 g/mol

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9.B PHARMACOKINETICS: METABOLISM *IN VIVO* (METABOLITE PATTERN IN THE MOUSE)

Test Article: Linagliptin

Location in Module 4: **4.2.2.4**
Report No.: **U05-2537**
Study No.: **A448/05BC**

Species / strain	Mouse	
Strain	Crl:CD1(ICR)	
Gender (M/F) / No. of animals:	M / 5 & F/ 5: urine and faeces sampling M / 3 & F/ 3: plasma and bile sampling	M / 4 & F/ 5: urine and faeces sampling
Feeding condition:	fasted	fasted
Vehicle / Formulation:	solution in water, acidified with hydrochloric acid	solution in saline, acidified with hydrochloric acid
Method of Administration:	oral via gavage / intraduodenal for bile sampling	intravenous bolus injection
Dose:	25 mg/kg	4 mg/kg
Radionuclide:	^{14}C	^{14}C
Specific Activity:	1.934 MBq/µmol (batch) about 0.12 MBq/µmol (dosed)	1.934 MBq/µmol (batch) about 1.14 MBq/µmol (dosed)

Additional information:
Mean data of male and female animals is presented since gender related differences of the metabolite pattern in urine, faeces, bile and plasma were not observed.

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9.B Pharmacokinetics: Metabolism *In Vivo* (Metabolite pattern in the mouse, Continued)

Report No.: U05-2537

	25 mg/kg p.o.								4 mg/kg i.v.		
compound code	plasma 1 h [nmol/l]	(% of sample radioactivity)	plasma 6 h [nmol/l]	(% of sample radioactivity)	urine 0 - 48 h [% of dose administered]	faeces 0 - 48 h [% of dose administered]	urine + faeces total [% of dose administered]	bile 0 - 6 h [% of dose administered]	urine 0 - 48 h [% of dose administered]	faeces 0 - 48 h [% of dose administered]	urine + faeces total [% of dose administered]
Linagliptin, BI 1356 (M473(1))	**16790**	**(74.2)**	**1337**	**(82.4)**	**18.3**	**48.4**	**66.7**	**4.5**	**22.8**	**45.3**	**68.1**
CD 1790 (M474(1))	372	(1.7)	22	(1.6)	-	2.9	2.9	-	-	1.7	1.7
M475(1)	-	-	-	-	-	0.5	0.5	-	-	1.2	1.2
M489(2)	441	(1.9)	22	(1.4)	0.4	1.8	2.2	0.3	0.3	1.3	1.6
M489(3) ,M491(1), M489(4)	656	(2.9)	78	(4.5)	0.3	3.5	3.9	0.7	0.4	4.5	4.9
M490(5)	790	(3.5)	32	(2.4)	-	1.7	1.7	-	-	1.7	1.7
M503(1)	-	-	-	-	-	0.8	0.8	-	-	1.0	1.0
M505(2)	-	-	-	-	-	3.1	3.1	0.3	-	3.7	3.7
M515(1)	-	-	-	-	-	1.3	1.3	-	-	1.0	1.0
M531(1), M506(1)	-	-	-	-	-	3.4	3.4	-	-	2.7	2.7
M610(1)	-	-	-	-	-	-	-	0.2	-	-	-
M636(1), M693(1)	610	(2.6)	38	(1.9)	0.5	-	0.5	1.5	0.4	-	0.4
M665(2)	1829	(7.7)	51	(4.3)	0.3	-	0.3	-	0.3	-	0.3
M812(1)	-	-	-	-	-	-	-	0.1	-	-	-
M812(2), M812(3)	-	-	-	-	0.2	-	0.2	-	0.3[a]	-	0.3[a]
m1	398	(1.7)	-	-	0.2	-	0.2	-	0.2	-	0.2
m2	735	(3.2)	40	(1.5)	0.3	-	0.3	-	0.4	-	0.4
m3	163	(0.7)	-	-	-	-	-	-	-	-	-
m4	-	-	-	-	-	0.3	0.3	-	0.2	0.3	0.5
m5	-	-	-	-	0.1	-	0.1	-	-	-	-
m6	-	-	-	-	< 0.1	-	< 0.1	-	-	-	-
m7	-	-	-	-	-	-	-	0.5	-	-	-
sum	22783	(100)	1620	(100)	20.6	67.7	88.3	8.0	25.3	64.5	89.9

empty cells: not detected, a: M812(2) only

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9.C PHARMACOKINETICS: METABOLISM *IN VIVO* (METABOLITE PATTERN IN THE CYNOMOLGUS MONKEY)

Test Article: Linagliptin

Location in Module 4: 4.2.2.4
Report No.: U06-2053
Study No.: A450/05BC

Species	Cynomolgus monkeys	
Gender (M/F) /	M / 3 & F/ 3: plasma, urine and faeces sampling	M / 3 & F/ 3: plasma, urine and faeces sampling
Number of animals	M / 2 & F/ 2: bile sampling	
Feeding condition:	fasted	fasted
Vehicle / Formulation:	solution in water, acidified with hydrochloric acid	solution in aqueous glucose (5%), acidified with hydrochloric acid
Method of Administration:	oral via gavage / intraduodenal for bile sampling	intravenous bolus injection
Dose (mg/kg):	5 mg/kg	1.5 mg/kg
Radionuclide:	^{14}C	^{14}C
Specific Activity:	about 0.24 MBq/µmol (dosed)	about 0.79 MBq/µmol (dosed)

Additional information:
Mean data of male and female animals is presented since gender related differences of the metabolite pattern in urine, faeces, bile and plasma were not observed.

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9.C Pharmacokinetics: Metabolism *In Vivo* (Metabolite pattern in the Cynomolgus monkey, Continued)

Report No.: U06-2053

	5 mg/kg p.o.						1.5 mg/kg i.v.				
compound code	plasma, 2 h [nmol/l] (% of sample radioactivity)		urine 0 - 48 h	faeces 0 - 72 h	urine + faeces total	bile 0 - 6 h	plasma, 1 h [nmol/l] (% of sample radioactivity)		urine 0 - 48 h	faeces 0 - 72 h	urine + faeces total
			[% of dose administered]						[% of dose administered]		
Linagliptin, BI 1356 (M473(1))	**816.6**	**(50.9)**	**8.4**	**11.3**	**19.7**	**7.9**	**590.5**	**(75.0)**	**12.8**	**17.9**	**30.8**
CD 1790 (M474(1))	486.1	(29.8)	-	0.4	0.4	-	144.6	(18.3)	-	-	-
M476(1)	-	-	-	1.4	1.4	-	-	-	-	-	-
M489(1)	-	-	-	1.8	1.8	0.3	-	-	-	2.1	2.1
M489(2)	-	-	-	0.6	0.6	0.3	-	-	-	2.0	2.0
M489(3)	-	-	-	2.0	2.0	0.3	-	-	-	2.6	2.6
M490(1,2,3), M531(2)	63.9[b]	(3.9)[b]	0.8	4.9	5.7	1.9	12.4[b]	(1.6)[b]	0.3	2.7	3.0
M490(4), M531(3)	40.1	(2.4)	-	-	-		-	-	-	-	-
M476(1), M504(2), M506(4)	110.1	(6.8)	-	6.5[c]	6.5[c]	1.2[c]	25.8	(3.3)	-	3.3 [c]	3.3 [c]
M505(1)	-	-	-	3.5	3.5	0.4	-	-	-	1.9	1.9
M506(1), M531(2), M650(1)	19.4[a]	(1.2)[a]	-	34.2	34.2	8.3	5.5[a]	(0.7)[a]	-	22.2	22.2
M507(1)	-	-	-	0.9	0.9	0.3	--	-	-	0.6	0.6
M515(1)	-	-	-	-	-	-	-	-	<0.1	-	<0.1
M522(1)	-	-	-	0.8	0.8	-	-	-	-	-	-
M636(3,4,5)	-	-	-	-	-	-	-	-	0.4	-	0.4
M665(2)	36.0	(2.3)	-	-	-	-	2.4	(0.3)	-	-	-
M693(1)	-	-	-	-	-	0.9	-	-	-	-	-
M707(1)	-	-	-	-	-	0.3	-	-	-	-	-
m1	-	-	0.5	-	0.5	-	-	-	0.3	-	0.3
m2	43.0	(2.7)	0.8	-	0.8	-	7.2	(0.9)	0.4	-	0.4
m6	-	-	0.1	-	0.1	-	-	-	-	-	-
sum	1615	(100.0)	10.6	68.2	78.8	22.1	788.4	(100.0)	14.3	55.3	69.6

- empty cells: not detected, [a]: M531(2) only, [b]: M490(1,2,3) only, [c]: M504(2) only

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9.D PHARMACOKINETICS: METABOLISM *IN VIVO* (METABOLITE PATTERN IN THE FEMALE RABBIT)

Test Article: Linagliptin

Location in Module 4: **4.2.2.4**
Report No.: **U06-2216**
Study No.: **A442/05BC**

Species	Rabbit
Strain	Crl:CHBB(HM)
Gender (M/F) / Number of animals	F / 3: urine and faeces sampling F / 3: plasma and bile sampling
Feeding condition:	fasted
Vehicle / Formulation:	solution in water, acidified with hydrochloric acid
Method of Administration:	oral via gavage / intraduodenal for bile and plasma sampling
Dose (mg/kg):	25 mg/kg
Radionuclide	^{14}C
Specific Activity	about 0.051 MBq/µmol (dosed)

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9.D Pharmacokinetics: Metabolism *In Vivo*
Metabolite pattern in the rabbit (Continued)

Report No.: U06-2216

compound code	25 mg/kg p.o.							
	plasma				urine	faeces	urine + faeces	bile
	1 h		4 h		0 - 48 h	0 - 72 h	total	0 - 4h
	[µmol/l] (% of sample radioactivity)				[% of dose administered]			
Linagliptin, BI 1356 (M473(1))	**12.6**	**(82.3)**	**7.91**	**(75.4)**	**15.6**	**41.9**	**57.5**	**1.1**
CD 1790 (M474(1))	0.42	(2.7)	0.39	(3.8)	-	0.4	0.4	-
M476(1)	-	-	-	-	0.1	-	0.1	-
M489(1)	0.43	(2.8)	0.59	(5.6)	0.7	18.6	19.4	2.5
M489(2)	0.29	(1.9)	-	-	-	-	-	-
M505(1)	-	-	-	-	-	0.6	0.6	0.1
M505(6), M665(2)	0.42	(2.7)	-	-	0.4	-	0.4	0.1
M507(1)	-	-	-	-	-	0.6	0.6	0.1
M610(2,3,4)	-	-	-	-	-	0.4	0.4	-
M650(1), M531(1)	0.32	(2.1)	-	-	0.5	1.3	1.8	0.1
M665(3), M665(4)	0.37	(2.4)	-	-	0.2	-	0.2	0.1
M693(1)	0.46	(3.0)	1.61	(15.3)	-	-	-	1.2
m1	-	-	-	-	<0.1	-	<0.1	-
m2	-	-	-	-	<0.1	-	<0.1	-
m3	-	-	-	-	<0.1	-	<0.1	-
m4	-	-	-	-	<0.1	-	<0.1	-
m5	-	-	-	-	<0.1	-	<0.1	-
m6	-	-	-	-	0.1	-	0.1	-
m7	-	-	-	-	<0.1	-	<0.1	-
m8	-	-	-	-	<0.1	-	<0.1	-
total	15.3	(100.0)	10.5	(100.0)	17.7	63.9	81.6	5.3

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9.E PHARMACOKINETICS: METABOLISM *IN VIVO* (METABOLITE PATTERN IN HEALTHY MALE VOLUNTEERS)

Test Article: Linagliptin

Location in Module 5: **5.3.2.2**
Report No.: **U07-1751**
Study No.: **A495/06BC**

Species	Healthy male subjects (Caucasian)	
Gender (M/F) / No. of subjects	M / 6: plasma, urine and faeces sampling	M / 6: plasma, urine and faeces sampling
Feeding condition:	fasted	fasted
Vehicle / Formulation:	solution aqueous tartaric acid (0.1%)	solution in acetate buffer
Method of Administration:	oral as drinking solution	intravenous infusion (1.5 h)
Dose:	10 mg/subject	5 mg/subject
Radionuclide	^{14}C	^{14}C
Specific Activity	0.1 MBq/µmol (dosed)	0.2 MBq/µmol (dosed)

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9.E Pharmacokinetics: Metabolism *In Vivo* (Metabolite pattern in healthy male volunteers, Continued)

Report No.: U07-1751

	10 mg/subject oral (trinking solution)				5 mg/subject intravenous infusion (1.5 h)					
compound code	1.5 + 3 + 6 h (sample pool) [nmol/l] (% of sample radioactivity)		urine 0 - 48 h	faeces 0 - 120 h	1.5 h [nmol/l] (% of sample radioactivity)		3 + 6 h (sample pool) [nmol/l] (% of sample radioactivity)		urine 0 - 48 h	faeces 0 - 240 h
Linagliptin, BI 1356 (M473(1))	**17.3**	**(74.2)**	**3.9**	**74.1**	**104.1**	**(93.5)**	**28.3**	**(83.9)**	**25.3**	**35.8**
CD 1790 (M474(1))	3.9	(16.9)	0.2	0.3	6.2	(5.6)	3.7	(10.9)	0.1	2.5
M476(1), M515(1)	-	-	0.1 [a]	-	-	-	-	-	0.1	-
M489(1)	0.1	(0.3)	0.2	4.5	-	-	0.2	(0.7)	0.3	9.3
M503(1)	-	-	<0.1	0.1	-	-	-	-	0.2	1.2
M504(2)	-	-	0.1	-	-	-	-	-	-	-
M531(2), M490(1), M506(1)	-	-	-	2.5	-	-	-	-	-	4.0 [b]
M620(1)	-	-	-	-	-	-	-	-	<0.1	-
M636(2)	-	-	-	-	-	-	-	-	0.1	-
M636(3), M636(4)	-	-	-	-	-	-	-	-	0.2	-
M650(1)	-	-	0.3	-	-	-	-	-	-	-
M665(3)	-	-	0.1	-	-	-	0.4	(1.2)	0.4	-
M665(8), M490(1)	-	-	0.3	-	-	-	-	-	-	-
M665(8), M531(1) M531(2)	-	-	-	-	-	-	-	-	0.7	-
M665(8), M650(1)	1.3	(5.5)	-	-	1.0	(0.9)	0.9	(2.7)	-	-
m1	-	-	0.1	-	-	-	-	-	<0.1	-
m2	-	-	0.2	-	-	-	-	-	0.1	-
m3	-	-	<0.1	-	-	-	-	-	<0.1	0.3
m4	0.7	(3.1)	-	-	-	-	0.2	(0.6)	-	-
m5	-	-	<0.1	-	-	-	-	-	<0.1	-
m6	-	-	<0.1	-	-	-	-	-	-	-
total	23.3	(100.0)	5.5	81.5	111.3	(100.0)	33.7	(100.0)	27.5	53.2

[a]: M515(1) only, [b]: M506(1) only, - empty cells: not detected

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9.F PHARMACOKINETICS: METABOLISM *IN VIVO* (METABOLITE PATTERN IN HUMANS)

Test Article: Linagliptin

Location in Module 5: **5.3.2.2**
Report No.: **U05-2659**
Study No.: **A439/05BC**

Species / strain	healthy male subjects / Caucasians of trial 1218.1
Gender (M/F) / No. of subjects	m (n = 6): plasma, urine sampling
Feeding condition:	fasted
Vehicle / Formulation:	tablet
Method of Administration:	oral
Dose:	25 and 600 mg/subject single dose

Additional information:
The metabolism of linagliptin was investigated in plasma (pool: 0.5 -8 h) and urine (pool: 0 – 24 h) samples of the clinical trail no. 1218.1 [U05-2072, Module 5.3.3.1] after a single oral dose of 25 mg and 600 mg, respectively. The search of linagliptin-metabolites in plasma and urine samples was performed by ESI-QTOF mass spectrometry using a modified high resolution parent ion discovery experiment. Due to the lack of synthetic reference compounds the relative metabolite concentrations were assessed by comparison of integral $[M+H]^+$ -ion intensities together with the simplified assumption of equimolar responses of different analytes. The inaccuracies of this approach are acknowledged however precise quantitative determination of coeluting metabolites could not be performed due to the absence of authentic reference material. However, relative metabolite concentrations were roughly estimated and classified into 3 groups: <1%, 1 – 10% and >10% of total drug related compounds (see table). CD 1790 was identified as dominant metabolite in human plasma.

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9.F Pharmacokinetics: Metabolism *In Vivo* (Metabolite pattern in humans, Continued)

Report No.: U05-2659

	25 mg/subject		600 mg/subject	
compound code	urine	plasma	urine	plasma
Linagliptin, BI 1356 (M473(1))	+++	+++	+++	+++
CD 1790 (M474(1))	++	+++	++	+++
M476(1)	++	++	++	+
M487(1)	+	-	++	+
M489(1)	++	+	++	++
M490(1)	++	+	++	+
M490(2)	+	+	++	+
M490(3)	+	+	++	+
M503(1)	+	-	++	+
M504(1)	-	-	++	-
M506(1)	++	+	++	+
M515(1)	+	-	++	+
M531(1)	-	-	++	+
M531(2)	+	-	++	+
M531(3)	-	-	++	+
M665(2)	+	-	++	++
M665(8)	+	-	++	++
M716(1)	+	-	++	-

+: traces: <1% of total drug related compounds (not depicted)
++: low: 1-10% of total drug related compounds
+++: high: >10% of total drug related compounds
-: not detected

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9.G PHARMACOKINETICS: METABOLISM IN VIVO

Test Article: Linagliptin

(INVESTIGATIONS ON NON-EXTRACTABLE RADIOACTIVITY IN PLASMA OF THE ^{14}C HUMAN ADME STUDY FOR THE QUANTITATIVE ASSESSMENT OF COVALENTLY BOUND DRUG-RELATED MATERIAL IN HUMAN PLASMA)

Location in Module 5: **5.3.2.2**
Report No.: **U08-1228**
Study No.: **A495/06BC**

Species	Healthy male subjects (Caucasian)	
Gender (M/F) / No. of subjects	M / 6: plasma, urine and faeces sampling	M / 6: plasma, urine and faeces sampling
Feeding condition:	fasted	fasted
Vehicle / Formulation:	solution aqueous tartaric acid (0.1%)	solution in acetate buffer
Method of Administration:	oral as drinking solution	intravenous infusion (1.5 h)
Dose:	10 mg/subject	5 mg/subject
Specific Activity	0.1 MBq/µmol (dosed)	0.2 MBq/µmol (dosed)

10 mg/subject p.o.				**5 mg/subject i.v.**			
sample pool	total radioactivity	non-extractable radioactivity		sample pool	total radioactivity	non-extractable radioactivity	
[h]	mean [nmol/l]	[%]	[nmol/l]	[h]	mean [nmol/l]	[%]	[nmol/l]
0.5 + 0.75	24.4	0.1	< 0.1	0.5 + 1	86.1	0.4	0.3
1 + 1.5	26.1	6.9	1.8	1.5 + 1.58 + 1.67 + 1.75	80.7	5.1	4.1
2 + 3	23.5	7.9	1.9	2 + 2.25 + 2.5	51.0	2.4	1.2
4 + 6	24.4	6.7	1.6	3 + 3.5 + 4	37.4	5.0	1.9
8 + 12	14.3	10.1	1.4	5 + 6	26.3	5.4	1.4
24 + 48 + 72	6.5	14.1	0.9	8 + 12	17.3	5.2	0.9
				24 + 48	8.6	9.4	0.8
				72 + 96 + 120	4.0	0.0	0.0

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9.G Pharmacokinetics: Metabolism In Vivo (Investigations on non-extractable radioactivity in plasma of the ^{14}C human ADME study for the quantitative assessment of covalently bound drug-related material in human plasma, Continued)

Report No.: **U08-1228**

Additional information:

This study was performed to obtain detailed quantitative data on the potential formation of covalently bound drug related material to plasma proteins after administration of linagliptin to humans. Sample material of the human ^{14}C-ADME trial was investigated [U07-1751, Module 5.3.2.2]. Due to the low specific radioactivity in combination with limited amounts of sample material, plasma samples were pooled of all 6 subjects and at least 2 sampling times. The sample material was processed by protein precipitation and exhaustively extraction of precipitates.

The concentrations of covalently bound drug related material to plasma proteins after both intravenous and oral administration were very low. In maximum they accounted for 4.1 nmol/l and 1.9 nmol/l after a single dose of 5 mg (i.v.) and 10 mg (p.o.), respectively. The data was in line with results the ^{14}C-ADME study in humans [U07-1751, Module 5.3.2.2]. Here the non-extractable part in plasma was calculated to 1.5% - 4.0% of sample radioactivity and was not further evaluated.

In conclusion, non-extractable radioactivity in plasma after both single intravenous and single oral administration of ^{14}C-linagliptin which may be due to covalently bound drug-related material was observed in humans. The amounts and concentrations were exceptional low and were considered to be neglectable.

Compared with data of ^{14}C-drug metabolism studies in animals, the rat was the only species with a pronounced formation of covalently bound radioactivity in plasma with >50% of sample radioactivity after 24 h post dose. However, the data demonstrated that animals of safety testing were adequately exposed to non-extractable radioactivity in plasma.

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9.G Pharmacokinetics: Metabolism In Vivo

Report No.: U08-1228

(Investigations on non-extractable radioactivity in plasma of the ^{14}C human ADME study for the quantitative assessment of covalently bound drug-related material in human plasma, Continued)

Summary table of non-extractable radioactivity in plasma of animal species following intravenous and oral administration of [^{14}C]linagliptin (means of males and females (rabbits: females only)).

species	Report No.	dose, route of admin.	sampling time [h]	total radioactivity [nmol/l]	non-extractable radioactivity [%]	non-extractable radioactivity [nmol/l]
rat	[U05-1981, Mod. 4.2.2.4]	1.0 mg/kg, i.v.	1	253	8.4	21.3
			3	182	21.3	38.8
		1.0 mg/kg, p.o.	1	184	12.5	23.0
			4	110	16.1	17.7
	[U08-1929, Mod. 4.2.2.5]	30.0 mg/kg, p.o.	24	37	52.1	19.4
mouse	[U05-2537, Mod. 4.2.2.4]	25.0 mg/kg, p.o.	1	22783	3.4	774.6
			6	1620	0.3	4.9
rabbit	[U06-2216, Mod. 4.2.2.4]	25.0 mg/kg, p.o.	1	15300	1.6	244.8
			4	10500	0.7	73.5
Cynomolgus monkey	[U06-2053, Mod. 4.2.2.4]	1.5 mg/kg, i.v.	1	788	5.1	40.2
		5.0 mg/kg, p.o.	2	1615	7.5	121.1
human	[U07-1751, Mod. 5.3.2.2]	5 mg/subject, i.v.	1.5	111	1.5	1.7
			3 + 6	34	3.0	1.0
		10 mg/subject, p.o.	1.5 + 3 + 6	23	4.0	0.9

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9.H PHARMACOKINETICS: METABOLISM IN VIVO (ENANTIOSELECTIVE ANALYSIS OF LINAGLIPTIN AND CD 1790 IN PLASMA OF HUMANS AND ANIMAL SPECIES)

Test Article: Linagliptin

Location in Module 5: **5.3.1.4**
Report No.: **U07-2471**
Study No.: **A478/06BC**

Method:
Linagliptin is a chiral compound with R-configuration at C-3 of the piperidinyl moiety. The major metabolite in humans and animal species was identified as the 3-hydroxy-piperidinyl derivative.
For the investigation of a potential chiral inversion of linagliptin and the determination of the absolute configuration of major metabolite, an enantioselective HPLC-MS/MS method was developed and validated with the reference compounds of known absolute configuration (linagliptin: R-configuration, BI 1355: S-configuration, CD 1789: R-configuration, CD 1790: S-configuration). Plasma samples after oral administration of linagliptin to humans and animal species were investigated.

Results:
In human plasma after a single oral administration of 600 mg linagliptin only the parent compound with R-configuration and the metabolite CD 1790 with S-configuration was identified. The corresponding enantiomers BI 1355 and CD 1789 were not detectable. With mean concentrations of linagliptin and CD 1790 at 1736 ng/ml and 110 ng/ml, and with limits of quantification of 0.23 ng/mL for BI 1355 BS and of 0.05 ng/mL for CD 1789, the plasma concentration of linagliptin and CD 1790 was at least 7500-fold and 2300-fold higher than the concentration of the enantiomers BI 1355 and CD 1789, respectively. These results demonstrate that there was negligible chiral inversion of linagliptin *in vivo* in humans, if it all present, and that the formation of the corresponding S-configured alcohol CD 1790 was highly stereo selective.
The enantiomeric excess of linagliptin and CD 1790 accounted for >99.9% in humans. In animal species the enantiomeric excess of linagliptin and CD 1790 was at least >98.2% and 87.3%, respectively.
It was concluded that all animal species of safety testing of linagliptin were exposed to CD 1790, the S-enantiomer which was formed in humans.

GLP compliance: no

Species	dose / route of administration	enantiomeric excess (e.e.), means [%]	
		linagliptin	CD 1790
Human	600 mg / p.o.	>99.9	>99.9
Cynomolgus monkey	5 mg/kg / p.o.	>99.8	>99.6
Mouse	100 mg/kg / p.o.	>99.9	>97.8
Rat	100 mg/kg / p.o.	>99.9	93.6
Rabbit	25 mg/kg / p.o.	98.2	87.3

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9.I PHARMACOKINETICS: METABOLISM *IN VIVO* (MECHANISM OF THE METABOLIC PATHWAY OF LINAGLIPTIN TO CD 1790)

Test Article: Linagliptin

Location in Module 4: 4.2.2.4
Report No.: U08-2140
Study No.: A515/07BC

Method:
The formation of CD 1790 was investigated in detail with human and rat liver microsomes, cytosol, human plasma and human blood with linagliptin and the ketone CD 10604 as starting material. For the identification of the reducing enzyme(s) incubation experiments in the pH-range from 5.5 to 7.4 together with the chemical inhibitors flufenamic acid, phenolphthalein, dexamethasone, medroxyprogesterone, quercitrin, rutine, ethanol and menadione were performed. Additional incorporation experiments with ^{18}O-labelled water were performed.

Results:
- CD 1790 was formed by a two step mechanism via the ketone CD 10604.
- The overall formation of CD 1790 BS was found to be CYP 3A4-dependent [U05-2525, Module 4.2.2.4]. The rate limiting part of the reaction was the oxidative desamination of linagliptin to the ketone CD 10604.
- The presence in vivo of CD 10604 in all animal species investigated was demonstrated with assistance of the reference material. The very low concentrations of CD 10604 indicated a very effective overall metabolic transformation of the parent drug to the alcohol CD 1790.
- CD 1790 with S-configuration was formed by reduction of CD 10604 with high stereo selectivity in humans and in all animal species investigated [U07-2471, Module 5.3.1.4].
- Incorporation experiments with O18-labeled water demonstrated the insertion of O18 into CD 1790 via the geminal diol of CD 10604.
- Investigations with subcellular fractions together with various chemical inhibitors at different pH-values indicated the predominant role of cytosolic aldo-keto reductase (AKR) enzymes in the reduction of CD 10604. A week contribution of carbonyl reductases (CR) could not be excluded. The lack of commercially available AKR isoforms and lack of specific AKR inhibitors did not allow the identification of individual AKR isoforms involved in the reduction of CD 10604.
- Liver was not the only site of CD 10604 reduction. CD 10604 reduction was also observed in human blood but not in human plasma.

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10. PHARMACOKINETICS: METABOLISM *IN VITRO*

Test Article: Linagliptin

Location in Module 4: 4.2.2.4
Report No.: U05-2525
Study No.: A168/04LU

Study system: Human liver microsomes, human kidney microsomes, human hepatocytes, expressed CYP enzymes
Test item: [^{14}C]linagliptin

Study performance:

Human liver and kidney microsomes (up to 2 mg microsomal protein/mL), and human expressed CYP enzymes were incubated with [^{14}C]linagliptin (50 µM) in 0.1 M Tris buffer pH 7.4 containing 5 mM magnesium chloride and NADPH (1 mM),up to 90 min at 37°C. The formation of metabolites was quantitatively assessed using HPLC with radioactivity detection. The effect of selective inhibitors on the in vitro metabolism by human liver microsomes was investigated in order to support the identification of the CYP enzymes that are involved in linagliptin metabolism.

Primary human hepatocytes (2.5*10^6 vital cells per well) were incubated at 37°C, 95% humidity, 10% CO2 and gently shaking. After a pre incubation period of 30 min, the substrate was added (final substrate concentrations: 5 µM and 50 µM).

Results:

The metabolism of [^{14}C]linagliptin by human liver microsomes and human hepatocytes was very low. The in vitro metabolism of [^{14}C]linagliptin by pooled human liver microsomes and human hepatocytes yielded the formation of two metabolites, metabolite U3 (oxidation in the quinazoline moiety) and CD 1790. CYP 3A4 was identified as the enzyme responsible for the formation of metabolite. The rates of metabolite formation were too low to calculate enzyme kinetic parameters. There were no indications for a contribution of other CYP enzymes in the metabolism of [^{14}C]linagliptin and there was no turnover of [^{14}C]linagliptin by human kidney microsomes.

time (incubation) [min]	**0**	**30**	**45**	**60**	**90**
substrate concentration [µM]	**50**	**50**	**50**	**50**	**50**
compounds [% of total radioactivity]	98.16	94.79	96.04	95.13	94.39
U3 (linagliptin oxidated in the quinazoline moiety)	ND	1.01	1.15	1.15	1.43
CD 1790	0.19	0.96	1.22	1.56	1.68

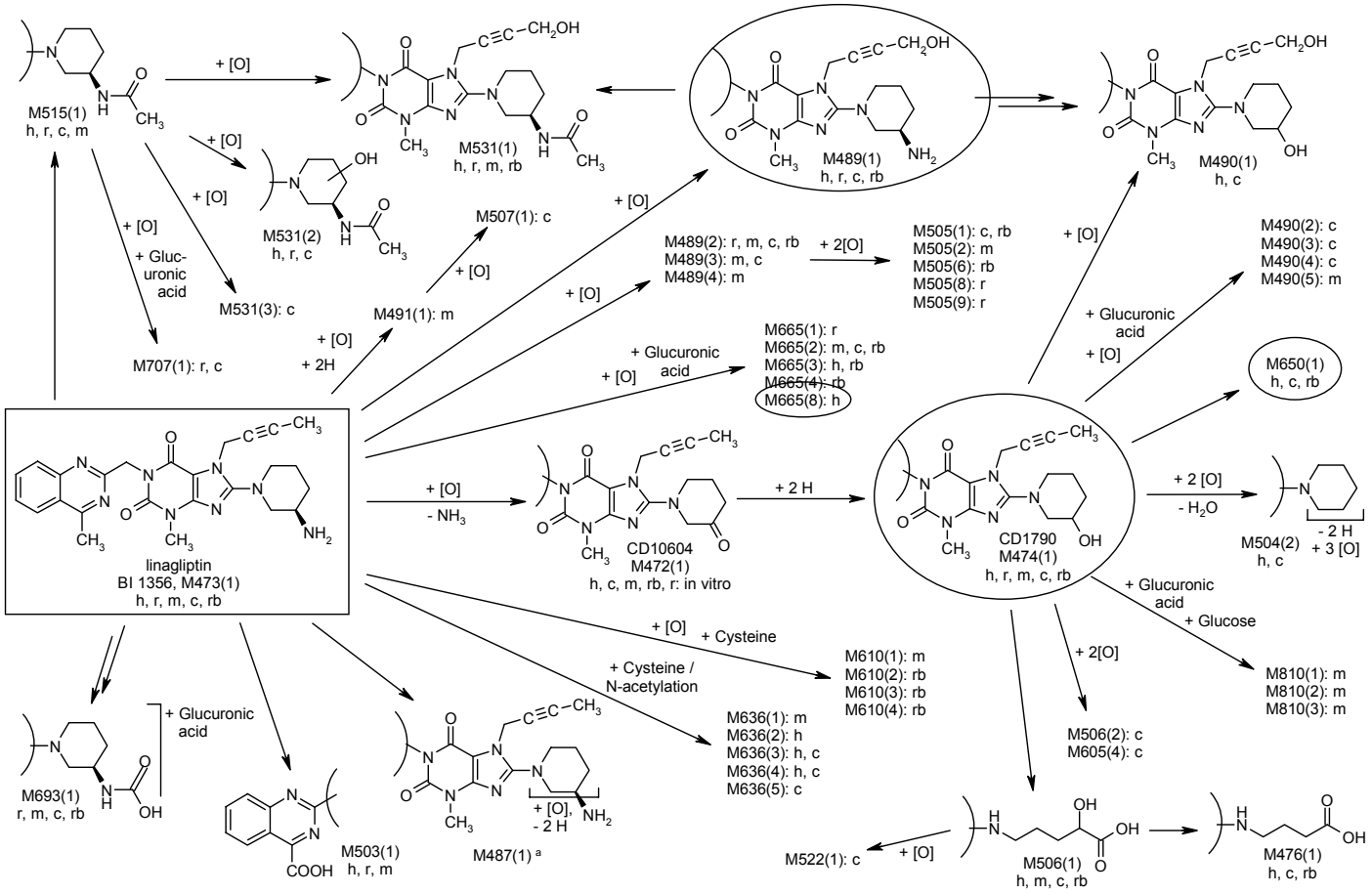
Additional Information:

Incubation experiments at 50 µM [^{14}C]linagliptin with human liver microsomes for 90 min in the presence of NADPH and with human hepatocytes for up to 24 h did not show covalent binding of [^{14}C]linagliptin derived material to human liver microsomal protein and human hepatocytes in the course of oxidative in vitro metabolism.

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11. PHARMACOKINETICS: METABOLIC PATHWAYS IN HUMANS AND ANIMAL SPECIES

Test Article: Linagliptin



Metabolism pathways of linagliptin (rectangle) in humans (h), rats (r), cynomolgus monkeys (c), mice (m) and rabbits (rb) after i.v. and oral administration. Human metabolites in plasma are indicated by a cycle. Additional metabolites which were not observed in humans but which were detected only in traces in animal species by mass spectrometry investigations are not depicted.

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12.A PHARMACOKINETICS: INDUCTION/INHIBITION OF DRUG METABOLIZING ENZYMES

Test Article: Linagliptin

Location in Module 4: **4.2.2.4**
Report No.: **U05-1012**
Study No.: **A161/04LU**

Type of study:	In vitro inhibition studies on cytochrome P450 dependent metabolic reactions
Test item:	Linagliptin

Method:
Experiments with pooled human liver microsomes in the presence of NADPH using test substrates for 10 human cytochrome P450 enzymes:

phenacetin O-deethylation	test for CYP 1A1/1A2;	coumarin 7-hydroxylation	test for CYP 2A6
S-mephenytoin N-demethylation	test for CYP 2B6;	paclitaxel 6α-hydroxylation	test for CYP 2C8
tolbutamide hydroxylation	test for CYP 2C9;	S-mephenytoin 4'-hydroxylation	test for CYP 2C19
bufuralol 1'-hydroxylation	test for CYP 2D6;	lauric acid 11-hydroxylation	test for CYP 2E1
nifedipine oxidation	test for CYP 3A;	testosterone 6ß-hydroxylation	test for CYP 3A4
erythromycin N-demethylation	test for CYP3A4;	lauric acid 12-hydroxylation	test for CYP 4A11
midazolam 1'-hydroxylation	test for CYP3A4;		

The test substrates were incubated in the presence of 0.1 µM, 1 µM, 10 µM and 100 µM of linagliptin. The potential in vitro inhibition of the selective test reactions was used to assess the potential for drug-drug interactions on comedications that may be caused by linagliptin.
In addition, the potential for mechanism-based inhibition of CYP 3A4 by linagliptin was investigated in human liver microsomes and/or expressed CYP enzymes. CYP 3A4 dependent nifedipine oxidation , erythromycin N-demethylation and testosterone 6β-hydroxylation was used as test assay.

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12.A Pharmacokinetics: Induction/Inhibition of Drug Metabolizing Enzymes (Continued)

Report No.: U05-1012

Results:

test reaction	CYP enzyme	IC_{50} [µM]	K_i [µM]
bufuralol 1'-hydroxylation	2D6	>100	ND
coumarin 7-□ydroxylation	2A6	>100	ND
erythromycin N-demethylation	3A4	41.6	ND
lauric acid 11-hydroxylation	2E1	>100	ND
lauric acid 12-hydroxylation	4A11	>100	ND
midazolam 1-hydroxylation	3A4	>100	ND
nifedipine oxidation	3A4	>100	ND
paclitaxel 6α-hydroxylation	2□8	>10	ND
phenacetin O-deethylation	1A2	>100	ND
S-mephenytoin N-demethylation	2B6	>100	ND
S-mephenytoin 4'-hydroxylation	2C19	>100	ND
testosterone 6ß-hydroxylation	3A4	36.3	115*
tolbutamide hydroxylation	2C9	>100	ND

ND = not determined

Mechanism-based inhibition:

test reaction	test system	K_I [µM]	k_{inact} [min^{-1}]
testosterone 6β-hydroxylation	expressed CYP 3A4	1.75	0.041
erythromycin N-demethylation	expressed CYP 3A4	6.0	0.027

Additional Information:
Linagliptin was considered to be a poor to moderate mechanism-based inhibitor of CYP 3A4. There was no inhibition of any of the other CYP isoenzymes that were investigated by this study.

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12.B PHARMACOKINETICS: INDUCTION/INHIBITION OF DRUG METABOLIZING ENZYMES

Test Article: Linagliptin

Location in Module 4: **4.2.2.4**
Report No.: **U09-1515**
Study No.: **A239/08LU**

Type of study: In vitro inhibition studies on cytochrome P450 dependent metabolic reactions
Test item: CD 1790

Method:
Experiments with pooled human liver microsomes in the presence of NADPH using test substrates for 10 human cytochrome P450 enzymes:

phenacetin O-deethylation	test for CYP 1A1/1A2;	coumarin 7-hydroxylation	test for CYP 2A6
bupropion hydroxylation	test for CYP 2B6;	amodiaquine *N*-deethylation	test for CYP 2C8
diclofenac 4'-hydroxylation	test for CYP 2C9;	flubiprofen 4'-hydroxylation	test for CYP 2C9
S-mephenytoin 4'-hydroxylation	test for CYP 2C19;	dextromethorphan *O*-deethylation	test for CYP 2D6
lauric acid 11-hydroxylation	test for CYP 2E1	nifedipine oxidation	test for CYP 3A4;
testosterone 6ß-hydroxylation	test for CYP 3A4	midazolam 1'-hydroxylation	test for CYP3A4;
lauric acid 12-hydroxylation	test for CYP 4A11		

The test substrates were incubated in the presence of 0.1 µM, 1 µM, 10 µM and 100 µM of CD 1790. The potential in vitro inhibition of the selective test reactions was used to assess the potential for drug-drug interactions on comedications that may be caused by CD 1790.
In addition, the potential for mechanism-based inhibition of CYP 3A4 by CD 1790 was investigated in human liver microsomes. CYP 3A4 dependent nifedipine oxidation, midazolam 1'-hydroxylation and testosterone 6β-hydroxylation was used as test assay.

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12.B Pharmacokinetics: Induction/Inhibition of Drug Metabolizing Enzymes (Continued)

Report No.: U09-1515

Results:

CYP enzyme	test reaction	IC_{50} [µM]	K_i [µM]	[I]/K_i	interaction potential
1A2	phenacetin *O*-deethylation	>100	ND	ND	remote
2A6	coumarin 7-hydroxylation	>100	ND	ND	remote
2B6	bupropion hydroxylation	>100	ND	ND	remote
2C8	amodiaquine *N*-deethylation	>100	ND	ND	remote
2C9	diclofenac 4'-hydroxylation	19.6	ND	ND	remote
2C9	flurbiprofen 4'-hydroxylation	15.2	8.28	0.0015	remote
2C19	*S*-mephenytoin 4'-hydroxylation	>100	ND	ND	remote
2D6	dextromethorphan *O*-deethylation	>100	ND	ND	remote
2E1	lauric acid 11-hydroxylation	>100	ND	ND	remote
3A4	nifedipine oxidation	>100	ND	ND	remote
3A4	testosterone 6β-hydroxylation	43.2	25.2	0.0005	remote
3A4	midazolam 1'-hydroxylation	>100	ND	ND	remote
4A11	lauric acid 12-hydroxylation	>100	ND	ND	remote

ND = not determined

Mechanism-based inhibition:

test reaction	test system	K_I [µM]	k_{inact} [min^{-1}]
testosterone 6β-hydroxylation	human liver microsomes	88.3	0.25

Additional Information:

CD 1790 was found to be a competitive inhibitor of CYP 2C9 and a mechanism-based inhibitor of CYP 3A4. By means of in vitro K_i-values (competitive inhibition), K_I and k_{inact}-values (mechanism-based inhibition) and in vivo therapeutic C_{max} values of the drug, an assessment of the in vivo effect was performed. In vivo plasma concentrations $C_{max,ss}$ of CD 1790 after multiple dosing of 10 mg linagliptin are in the nM range $C_{max,ss}$: 12.6 nM, after multiple dosing of 10 mg in Japanese, (U08-3213, Module 5.3.3.2). $C_{max,ss}$: 2 - 3 nM, after multiple dosing of 5 mg in Caucasian, study No. (U09-1247, Module 5.3.3.4).

An assessment of the interaction potential of CD 1790 by mechanism-based inhibition of CYP 3A4 resulted in an estimated 1.1-fold decrease in intrinsic clearance. Therefore, it was considered unlikely that inhibition of CYP isoenzymes under these in vivo conditions would occur.

Thus a relevant effect of CD 1790 on the CYP catalysed metabolism of co-medicated drugs is unlikely.

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13.A PHARMACOKINETICS: EXCRETION

Test Article: Linagliptin

Location in Module 4: 4.2.2.5
Report No.: U05-2607
Study No.: A034/05FU

Species / Strain	Mouse	
Strain	CD-1	
Gender (M/F) / Number of animals	M / 5 & F /5	M / 5 & F /5
Feeding condition	ad libitum	ad libitum
Vehicle/Formulation	solution in tap water	solution in isotonic sodium chloride
Method of Administration	oral	intravenous
Dose (mg/kg)	25 mg/kg	4 mg/kg
specific radioactivity	about 0.12 MBq/µmol	about 1.14 MBq/µmol
Analyte	radioactivity	radioactivity
Assay	LSC	LSC

Excretion route (data given as % of dose)	time [h]	Urine	Faeces	time [h]	Urine	Faeces
	0 - 8	7.3	NA	0 - 8	19.4	NA
	0 - 24	19.9	57.4	0 - 24	25.0	58.5
	0 - 48	20.6	67.7	0 - 48	25.3	64.5
	0 - 72	20.6	68.8	0 - 72	25.5	65.2
	0 - 96	20.7	69.1	0 - 96	25.5	65.6
	0 - 175	20.7	69.2	0 - 175	25.6	66.1
total (including cage wash)	0 - 175	91.4		0 - 175	93.5	
biliary excretion 0 - 6 h (maximum value from 1 animal)	0 - 6	46.2		not performed		

NA = not applicable

Additional Information:

After oral and intravenous administration of [^{14}C]linagliptin the major route of excretion in mice is by bile and faeces. No relevant gender differences were observed.

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13.B PHARMACOKINETICS: EXCRETION

Test Article: Linagliptin

Location in Module 4: 4.2.2.3
Report No.: U08-1364
Study No.: A084/07FU

Species	Mouse
Strain	DPP-4 knockout (C57BL/6TgH(CD26)-CIML) wildtype (C57BL/6J)
Gender (M/F) / Number of animals	F / 3 per strain and dose groups, i.e. 36 females in total
Feeding condition	ad libitum
Vehicle/Formulation	solution in saline
Method of Administration	intravenous (bolus) injection
Dose range (mg/kg)	0.01 - 10
specific radioactivity	2.14 - 0.214 MBq/µmol
Analyte	radioactivity
Assay	LSC

Urinary excretion of radioactivity
0 – 72 h after dosing
[% of dose]

Dose [mg/kg]	**DPP-4 knockout**	**wildtype**
0.01	24.2	2.7
0.1	24.7	12.7
0.3	20.3	20.1
1	16.8	16.7
3	19.4	22.8
10	22.8	24.5

Additional Information:
The renal excretion of radioactivity was dose-dependent in female wildtype mice, whereas in DPP-IV knockout mice a virtually constant renal excretion was observed between 0.01 and 10 mg/kg. Taken together, non-linear pharmacokinetics were observed in wildtype mice and were absent in DPP-IV knockout mice, demonstrating that the non-linearity at low dose levels is exclusively attributed to DPP-IV in wildtype mice.

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13.C PHARMACOKINETICS: EXCRETION

Test Article: Linagliptin

Location in Module 4: 4.2.2.5
Report No.: U04-1731
Study No.: A017/04FU

Species	Rat	
Strain	Crl:WI(Han)	
Gender (M/F) / Number of animals	M / 5 & F / 5 (M/ 4 for bile)	M / 5 & F / 5 (M/ 4 for bile)
Feeding condition	fasted: from 18 h before to 2h after administration	fasted: from 18 h before to 2h after administration
Vehicle/Formulation	solution in tap water	solution in isotonic sodium chloride
Method of Administration	oral	intravenous
Dose (mg/kg)	1	1
specific radioactivity	about 0.65 MBq/µmol	about 0.65 MBq/µmol
Analyte	radioactivity	radioactivity
Assay	LSC	LSC
Excretion route		

	1 mg/kg p.o.					1 mg/kg i.v.				
	radioactivity (% of dose)					radioactivity (% of dose)				
time interval [h]	**Urine**	**Faeces**	**Total (without carcass)**	**Bile**	**Carcass**	**Urine**	**Faeces**	**Total (without carcass)**	**bile**	**Carcass**
0 - 8 h	0.5	NA	0.5	NA	NA	17.7	-	17.7	NA	NA
0 - 24 h	1.3	83.1	84.5	NA	NA	19.5	65.6	85.8	NA	NA
0 - 48 h	1.4	93.2	94.8	NA	NA	19.6	73.1	93.6	NA	NA
0 - 72 h	1.5	94.8	96.5	NA	NA	21.6	72.3	94.8	NA	NA
0 - 96 h	1.6	95.1	97.0	NA	4	21.7	72.8	95.4	NA	4.8
0 - 6 h	NA	NA	NA	37.5	NA	NA	NA	NA	42.8	NA

NA = not applicable

Additional Information:

After oral and intravenous administration of [^{14}C]linagliptin the major route of excretion in mice is by bile and faeces. No relevant gender differences were observed.

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13.D PHARMACOKINETICS: EXCRETION

Test Article: Linagliptin

Location in Module 4: 4.2.2.5
Report No.: U09-1831
Study No.: A177/07RB, A183/08RB

Species / Strain	Rat
Strain	Crl:WI(Han)
Gender (M/F) / Number of animals	M / 4 per group
Feeding condition	ad libitum
Vehicle/Formulation	solution in isotonic sodium chloride
Method of Administration	i.v. bolus
Dose	1 mg/kg
Analyte	linagliptin
Assay	HPLC-MS/MS

			Group 1 (activated charcoal)				Group 2 (control)			
Parameter	Unit	Gender	N	mean	CV (%)	SD	N	mean	CV (%)	SD
C_{max}	[nmol/L]	m	4	987	10.6	104	4	936	42.8	400
$t_{1/2}$	[h]	m	3	38.4	14.3	5.49	4	49.9	18.1	9.03
$AUC_{0\text{-}inf}$	[(nmol·h)/L]	m	3	1040	10.8	112	4	1410	24.1	342
CL	[mL/(min*kg)]	m	3	34.2	11.4	3.90	4	26.2	25.2	6.58
MRT_{disp}	[h]	m	3	21.6	16.6	3.60	4	33.6	31.2	10.5
V_{ss}	[L/kg]	m	3	43.9	9.6	4.21	4	50.3	16.4	8.27

Additional Information:
The plasma level time course of linagliptin and the derived pharmacokinetic parameters were very similar in both groups.

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13.D Pharmacokinetics: Excretion (Continued)

Report No.: U09-1831

Species / Strain	Rat		
Strain	Crl:WI(Han)		
Gender (M/F) / Number of animals	M / 4	M / 4	M / 4
Feeding condition	ad libitum	ad libitum	ad libitum
Vehicle/Formulation	solution in tap water	donor bile from rats dosed i.v. with [^{14}C]linagliptin	spiked bile
Method of Administration	intraduodenal	intraduodenal	intraduodenal
Dose (mg/kg)	1	approx. 1	1
specific radioactivity	about 0.5 MBq/µmol	about 0.5 MBq/µmol	about 0.5 MBq/µmol
Analyte	radioactivity	radioactivity	radioactivity
Assay	LSC	LSC	LSC
Excretion route	bile	bile	bile
Time **0 - 6 h**	30.5 ± 9.2% of dose	6.7 ± 5.4% of dose	2.6 ± 1.1% of dose

Additional Information:
It can be concluded, that the enterohepatic circulation of linagliptin in rats is about 15% and highly variable at a dose of 1 mg/kg. Furthermore, bile constituents have a negative effect on the intestinal absorption of linagliptin, explaining the unexpectedly low fraction of linagliptin which undergoes an enterohepatic cycle.

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13.E PHARMACOKINETICS: EXCRETION

Test Article: Linagliptin

Location in Module 4: 4.2.2.5
Report No.: U09-2475
Study No.: A095/08FU

Species	Rat			
Strain	wildtype Fischer rat (F344/DuCrl)			
Gender (M/F) / Number of animals	M / 4	M / 4	M / 4	M / 4
Feeding condition	reduced food before dosing, afterwards ad libitum	reduced food before dosing, afterwards ad libitum	reduced food before dosing, afterwards ad libitum	reduced food before dosing, afterwards ad libitum
Vehicle/Formulation	solution in tap water	solution in tap water	solution in tap water	solution in tap water
Method of Administration	p.o.	p.o.	p.o.	p.o.
Dose (mg/kg)	1	2.5	10	30
specific radioactivity	2.138 MBq/µmol	about 0.9 MBq/µmol	about 0.2 MBq/µmol	about 0.07 MBq/µmol
Analyte	radioactivity and linagliptin	radioactivity and linagliptin	radioactivity and linagliptin	radioactivity and linagliptin
Assay	LSC / HPLC-MS/MS	LSC / HPLC-MS/MS	LSC / HPLC-MS/MS	LSC / HPLC-MS/MS

Dose [mg/kg]	mean linagliptin in urine [% of dose]	SD	mean radioactivity in urine [% of dose]	SD
1.0	0.2	0.3	0.7*	0.4*
2.5	6.1	0.9	5.7	1.1
10.0	11.7	1.7	13.1	2.2
30.0	19.6	4.1	22.1	3.4

* N=3

Additional Information:
Renal elimination of linagliptin after oral administration is strongly dose dependent in rats. Urinary excretion of parent compound linagliptin is by far dominating at dose levels above 1 mg/kg and metabolites play a negligible role in renal elimination of linagliptin in rats.

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13.F PHARMACOKINETICS: EXCRETION

Test Article: Linagliptin

Location in Module 4: 4.2.2.5
Report No.: U09-2339
Study No.: A095/08FU, A585/08BC

Species	Rat	
Strain	wildtype Fischer rat (Crl:F344/DuCrl)	
Gender (M/F) / Number of animals	M / 6 per group	
Feeding condition	fasted	
Vehicle/Formulation	solution in acetic acid/sodium acetate buffer	
Method of Administration	i.v. bolus of [^{14}C]linagliptin	
Dose (mg/kg)	15	15
Co-administration	i.v. infusion of 7.5 mg/kg/h Zosuquidar for 6.5 h, starting 30 min before [^{14}C]linagliptin dosing	i.v. infusion of vehicle for 6.5 h, starting 30 min before [^{14}C]linagliptin dosing
specific radioactivity	about 0.04 MBq/µmol	about 0.04 MBq/µmol
Analyte	radioactivity and linagliptin	radioactivity and linagliptin
Assay	LSC / HPLC-MS/MS	LSC / HPLC-MS/MS

	Zosuquidar group			**Vehicle group**		
time interval	biliary excretion of radioactivity [nmol ^{14}C]	linagliptin [% of sample radioactivity]	biliary excretion of linagliptin [nmol]	biliary excretion of radioactivity [nmol ^{14}C]	linagliptin [% of sample radioactivity]	biliary excretion of linagliptin [nmol]
0 - 1 h	657.0	12.95	85.1	880.2	21.27	187.2
1 – 2 h	616.3	10.70	65.9	715.1	21.36	152.7
2 - 3 h	501.7	5.78	29.0	547.4	20.11	110.1
3 - 4 h	443.9	5.65	25.1	386.6	20.54	79.4
4 – 5 h	385.7	6.04	23.3	299.5	23.30	69.8
5 - 6 h	418.0	5.72	23.9	203.4	26.13	53.2
Total (0 - 6 h)	3022.7		252.3	3032.2		652.4
% of dose	34.0#		3.2	38.0#		8.2
Average Biliary excretion rate within 6 hours [nmol/min]	**8.4**		**0.7**	**8.4**		**1.8**

mean of individual data. Note, that all other data are derived from pooled bile samples.

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13.F Pharmacokinetics: Excretion (Continued)

Report No.: U09-2339

Additional Information:	Biliary elimination of parent compound and metabolites plays a central role in the disposition of linagliptin. Furthermore, parent compound in bile originates to a significant part from an active transport via P-gp. The biliary excretion of parent drug within 6 hours was significantly reduced by P-gp inhibition. The terminal biliary excretion half-life of both parent drug as well as metabolites was reduced by P-gp inhibition: In the vehicle group the estimated half-life of parent drug/metabolites was 2.3/2.7 hours, whereas in the Zosuquidar group the respective half life was 11/7.5 hours. Overall the data indicate that the metabolic clearance followed by biliary excretion is able to compensate for the decreased elimination of parent compound.

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13.G PHARMACOKINETICS: EXCRETION

Test Article: Linagliptin

Location in Module 4: **4.2.2.3**
Report No.: **U08-1540**
Study No.: **B84497**

Species:	Rat
Strain	(Wistar, HanRcc:WIST)
Gender (M/F) / Number of animals:	M / 15 & F / 9
Feeding condition:	ad libitum
Vehicle/Formulation:	solution in tap water
Method of Administration:	oral (gavage), once daily
Dose:	2 mg/kg
Radionuclide:	^{14}C
Specific Activity:	about 0.24 MBq/µmol
Sampling time:	once daily for 14 days

Study design:

The disposition of linagliptin was investigated in male and female rats after repeated oral administration of ^{14}C labelled linagliptin over 14 consecutive days. 24 h after the 1st, 4th, 7th, 10th and 14th dose, subgroups of 3 animals were sacrificed and tissue concentrations of radioactivity were determined (see 2.6.5.5). Urine and faeces of one subgroup of males were collected in daily intervals in order to evaluate the excretion profile during and after a multiple dosing regimen and the recovery of the administered radioactivity.

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13.G Pharmacokinetics: Excretion (Continued)

Report No.: U08-1540

	radioactivity in urine and faeces % of total dose			
interval	**Urine**		**Faeces**	
(days)	Mean	SD	Mean	SD
0 - 1	**0.03**	0.01	**5.55**	0.70
1 - 2	**0.08**	0.05	**5.50**	0.42
2 - 3	**0.07**	0.02	**5.94**	0.74
3 - 4	**0.06**	0.01	**5.71**	0.15
4 - 5	**0.06**	0.01	**6.24**	0.12
5 - 6	**0.06**	0.01	**6.24**	0.06
6 - 7	**0.08**	0.02	**6.91**	0.31
7 - 8	**0.08**	0.01	**6.78**	0.68
8 - 9	**0.07**	0.01	**6.84**	0.44
9 - 10	**0.08**	0.02	**6.71**	0.22
10 - 11	**0.08**	0.02	**7.20**	0.13
11 - 12	**0.07**	0.02	**6.95**	0.19
12 - 13	**0.07**	0.01	**6.89**	0.16
13 - 14	**0.07**	0	**6.43**	0.47
Subtotal	**0.96**	0.18	**89.89**	1.64

Total excretion (incl. cage wash): 91.15% of total dose
Tissue residues : 0.56% of total dose
Total recovery: 91.71% of total dose

Additional Information:
Steady state based on excretion data was reached quickly within 2 days after oral dosing of 2 mg/kg [^{14}C]linagliptin. Overall, the major elimination route is by faeces with about 90% of the dose, whereas renal excretion accounts for an almost negligible fraction of about 1 % of the total dose.

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13.H PHARMACOKINETICS: EXCRETION

Test Article: Linagliptin

Location in Module 4: 4.2.2.5
Report No.: U10-2457
Study No.: A209/09RB

Species	Rat
Strain	Crl:WI(Han)
Gender (M/F) / Number of animals	M / 6
Feeding condition	ad libitum
Vehicle/Formulation	solution in acetic acid/sodium acetate buffer
Method of Administration	i.v. bolus of [^{14}C]linagliptin
Dose (mg/kg)	1
specific radioactivity	2.069 MBq/µmol
Analyte	radioactivity
Assay	LSC

Matrix	**Time period [h]**	**Excretion of radioactivity [% of dose]**
urine	0 - 8	19.9
	0 - 24	23.2
urine (incl. cage wash)	**0 - 24**	**28.3**
feces	0 - 24	2.1
feces and gut content	**0 - 24**	**11.7**
bile	**0 - 24**	**49.0**
total recovery	**0 - 24**	**89.1**

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13.I PHARMACOKINETICS: EXCRETION

Test Article: Linagliptin

Location in Module 4: 4.2.2.5
Report No.: U06-1087
Study No.: A032/05FU

Species	Rabbit	
Strain	Crl:CHBB(HM)	
Gender (M/F) / Number of animals	F / 3	
Feeding condition	ad libitum	
Vehicle/Formulation	acidified solution in tap water	
Method of Administration	oral (gavage)	
Dose (mg/kg)	25	
specific radioactivity	about 0.051 MBq/µmol	
Analyte	radioactivity	linagliptin
Assay	LSC	HPLC MS/MS
Excretion route		

Time interval [h]	Urine (radioactivity) [% of dose]	Urine (linagliptin) [% of dose]	Faeces [% of dose]	Total [% of dose]	Bile (radioactivity) [% of dose]
0 - 8 h	11.1	8.6	-	-	-
0 - 24 h	17.3	13.5	37.1	-	-
0 - 48 h	17.7	13.6	60	-	-
0 - 72 h	18	13.6	63.9	-	-
0 - 96 h	18.1	13.6	66.7	86.5	-
0 - 4 h	-		--	-	5.4*

* maximum of 1 out of 3 rabbits where bile sampling was performed for in vivo metabolism investigations

Additional Information:
Exploratory autoradiography was performed wit the kidneys of these animals, demonstrating a long retention time of linagliptin related radioactivity in the renal cortex (cf. 2.6.5.5). In addition, radioactivity concentrations 7 days after dosing indicated also long residence time in the liver.

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13.J PHARMACOKINETICS: EXCRETION

Test Article: Linagliptin

Location in Module 4: 4.2.2.2 / 4.2.2.5
Report No.: U06-1132 / U05-2668
Study No.: AA27415

Species	Cynomolgus monkey		
Gender (M/F) / Number of animals	M / 3 & F / 3	M / 3 & F / 3	M / 2 & F / 2
Feeding condition	ad libitum		
Vehicle/Formulation	solution in tap water	solution in 5 % glucose	solution in tap water
Method of Administration	oral (gavage)	intravenous (bolus)	intraduodenal (under terminal anaesthesia)
Dose (mg/kg)	5 mg/kg	1.5	5
specific radioactivity	about 0.24 MBq/µmol	about 0.79 MBq/µmol	about 0.24 MBq/µmol
Analyte	radioactivity / linagliptin	radioactivity / linagliptin	radioactivity / linagliptin
Assay	LSC / HPLC-MS/MS	LSC / HPLC-MS/MS	LSC / HPLC-MS/MS
Excretion route			

	5 mg/kg p.o.			**1.5 mg/kg i.v.**			**5 mg/kg i.d.**
Time [h]	Urine (radioact.)	Urine (parent)	Faeces (radioact.)	Urine (radioact.)	Urine (parent)	Faeces (radioact.)	Bile (radioact.)
	[% of dose]			[% of dose]			[% of dose]
0 - 8	8.4	4.2	-	10.2	7.7	-	27.1 (0-6h)
0 - 24	9.8	4.9	39.4	13.7	10.2	39.4	ND
0 - 48	10.6	5.2	65.2	14.3	10.6	53.8	ND
0 - 72	10.7	5.2	68.2	14.7	10.9	55.3	ND
0 - 96	10.9	5.2	68.8	14.9	11.0	55.7	ND
0 - 120	11.0	5.3	69.0	15.0	11.1	56.1	ND
0 - 144	11.0	5.3	69.2	15.2	11.1	56.4	ND
0 - 168	11.1	5.3	70.0	15.3	11.2	56.6	ND

ND = not determined

Additional Information:
Exploratory autoradiography was performed wit the kidneys of these animals, demonstrating a long retention time of linagliptin related radioactivity in the renal cortex (cf. 2.6.5.5) . In addition, radioactivity concentrations 7 days after dosing indicated also long residence time in the liver. Renal clearance decreased with decreasing plasma concentrations of linagliptin, indicating non-linear mechanisms in the renal excretion of linagliptin.

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14. PHARMACOKINETICS: EXCRETION IN TO BILE

Test Article: Linagliptin

Not applicable

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15.A PHARMACOKINETICS: DRUG - DRUG INTERACTIONS INDUCTION OF CYTOCHROME P450 IN HUMAN HEPATOCYTES

Test Article: Linagliptin

Location in Module 4: 5.3.2.2
Report No.: U08-1198
Study No.: 300866881

Type of study: Assessment of potential induction of cytochrome P450 enzymes in cultured human hepatocytes
Test item: linagliptin

Method: The induction potential of cytochrome P450 (CYP) isoforms by linagliptin in fresh cultured human hepatocytes was measured by in situ catalytic activity assays selective for CYP1A2, CYP2B6 and CYP3A4. Hepatocytes were used of three individual tissue donors, linagliptin was incubated for three days at concentrations of 0.001, 0.005, 0.02, 0.2 and 2 µM. Prototypical inducers (ß-naphthofalvone, phenobarbital, rifampicin) served as positive controls.

Results: No induction of > 40% of the positive control inducer occurred for any of the selected enzymes tested in three donors. Based on these criteria, it can be concluded that BI1356 BS is not an inducer of CYP1A2, CYP2B6 and CYP3A4. At the highest concentration that was investigated (2 µM) modest decreases of enzyme activities CYP1A2, CYP2B6 and CYP3A4 were observed.

Tabulated results: Concentration-dependent effects on enzyme activity in cultured human hepatocytes for CYP1A2, CYP2B6 and CYP3A4 that were assayed by using phenacetin O-deethylation, mephenytoin N-demethylation and testosterone 6ß-hydroxylation, respectively. Displayed are the arithmetic mean (N = 3, arithmetic coefficients of variation (%) are given in brackets) changes compared to the negative control experiments.

treatment	conc. [µM]	x-fold induction (control =1) CYP1A2	CYP2B6	CYP3A
linagliptin	0 (control)			
linagliptin	0.001	0.99 (10.3)	0.84 (11.0)	1.04 (13.2)
linagliptin	0.005	0.97 (3.2)	0.97 (31.0)	0.92 (2.3)
linagliptin	0.02	0.84 (0.7)	0.96 (22.3)	0.97 (11.5)
linagliptin	0.2	0.84 (1.8)	0.88 (34.3)	0.83 (10.8)
linagliptin	2	0.77 (9.1)	0.81 (31.1)	0.46 (25.9)
ß-naphthoflavone	20	10.77 (28.3)	ND	ND
phenobarbital	2000	ND	14.70 (77.9)	ND
rifampicin	10	ND	ND	14.83 (72.6)

ND = not determined

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15.B PHARMACOKINETICS: DRUG - DRUG INTERACTIONS ENZYME INDUCTION IN RAT

Test Article: Linagliptin

Location in Module 4: 4.2.2.4
Report No. U04-2193
Study No. DODF1012

Type of study: Enzyme induction study in the rat in vivo
Test item: linagliptin

Method:
Groups of 5 male Wistar rats (strain CrlGlxBrlHan:WI) received a daily oral dose for a period of 4 days of either 6 mg/kg or 60 mg/kg of linagliptin suspended in 0.5% Natrosol. The prototype inducers, phenobarbital (120 mg/kg p.o.), ß-naphthoflavone (20 mg/kg i.p.), dexamethasone (100 mg/kg i.p.) and clofibric acid (200 mg/kg p.o.) were administered according to the same time schedule and served as positive controls, vehicle-treated animals served as negative controls. The animals were sacrificed on day five. As indicators of a CYP induction were determined: liver weight (relative to body weight), microsomal protein content (per gram liver tissue), microsomal CYP content relative to microsomal protein (total CYP content), enzyme activities for ethoxyresorufin-O-dealkylation (EROD, CYP1A, "methylcholanthrene-type" induction), pentoxyresorufin-O-dealkylation (PROD, CYP2B, "phenobarbital-type" induction), testosterone 6ß-hydroxylation (T-6ß-OH CYP3A, "dexamethasone-type" induction) and lauric acid 11- and 12-hydroxylation (L-11-OH, L-12-OH, CYP4A, peroxisome proliferator induction and CYP2E1 induction).

Results:
Oral dosing of linagliptin to five male Wistar rats once daily for 4 days caused no induction of CYP1A and CYP3A activities and had no effect on the liver to body weight ratios, total amount of microsomal protein or total amount of hepatic cytochrome P450. CYP4A and CYP2E1 enzyme activities were slightly elevated (1.22-fold) at the 6 mg/kg dose, which was considered as being of no relevance as there was no significant change at the high dose level of 60 mg/kg. CYP2B showed a minor but significant reduction (0.68-fold) in the 60 mg/kg dose group but no elevated enzyme activity by induction of CYP2B enzymes.

Tabulated results:

Factor of increase (control group = factor 1)								
treatment	**liver weight**	**microsomal protein**	**total CYP**	**EROD**	**PROD**	**T-6ß-OH**	**L-11-OH**	**L-12-OH**
6 mg/kg linagliptin	0.940	0.962	1.06	0.985	0.773	1.19	1.22*	1.34*
60 mg/kg linagliptin	1.01	1.14	0.975	0.997	0.680*	1.16	1.12	1.23
120 mg/kg phenobarbital	0.973	2.02*	5.47***	11.1***	377***	6.16***	2.48***	2.13***
20 mg/kg ß-naphthoflavone	0.840*	1.42	2.67***	104***	3.35*	2.64**	1.26*	1.82***
100 mg/kg dexamethasone	1.34**	1.99**	7.43***	2.51**	12.7***	6.67***	1.55***	3.32***
200 mg/kg clofibric acid	0.969	2.60***	2.31**	3.49***	19.4***	3.16***	3.63***	19.2***

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

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15. C PHARMACOKINETICS: DRUG - DRUG INTERACTIONS IN VITRO INHIBITION OF CYTOCHROME P450 ENZYMES

Test Article: Metformin mono

Location in Module: 4.2.2.4
Report No. U08-2174
Study No. A244/08LU

Type of study: **In vitro inhibition of cytochrome P450 enzymes, in vitro drug-drug interaction study**
Test item: Metformin hydrochloride (DI 618105 CL)

Method:
Experiments with pooled human liver microsomes in the presence of NADPH using test substrates for 10 human cytochrome P450 enzymes:

phenacetin O-dealkylation	test for CYP 1A1/1A2;	coumarin 7-hydroxylation	test for CYP 2A6
bupropion hydroxylation	test for CYP 2B6;	amodiaquine N-deethylation	test for CYP 2C8
diclofenac 4'-hydroxylation	test for CYP 2C9;	flurbiprofen 4'-hydroxylation	test for CYP 2C9
S-mephenytoin 4'-hydroxylation	test for CYP 2C19	dextromethorphan O-deethylation	test for CYP 2D6;
lauric acid 11-hydroxylation	test for CYP 2E1	nifedipine oxidation	test for CYP 3A;
testosterone 6β-hydroxylation	test for CYP 3A	midazolam 1'-hydroxylation	test for CYP 3A
lauric acid 12-hydroxylation	test for CYP 4A11		

The test substrates were incubated in the presence of 0.1 µM, 1 µM, 10 µM and 100 µM of metformin. The potential in vitro inhibition of the selective test reactions was used to assess the potential for drug-drug interactions on comedications that may be caused by metformin.
In addition, the potential for mechanism-based inhibition of CYP 3A4 by metformin was investigated. CYP 3A4 dependent nifedipine oxidation, testosterone 6β-hydroxylation and midazolam 1'-hydroxylation were used as test assays.

Tabulated results:
No pronounced inhibition of the various test reactions by metformin hydrochloride was observed. There was no indication of a mechanism-based inhibition of CYP 3A4 by metformin hydrochloride.

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15. C PHARMACOKINETICS: DRUG - DRUG INTERACTIONS
In vitro inhibition of cytochrome P450 enzymes (Continued)

Report No. **U08-2174**

Tabulated results:

test reaction	CYP enzyme	IC_{50} [µM]	K_i [µM]
phenacetin O-dealkylation	1A1/1A2	> 100	ND
coumarin 7-hydroxylation	2A6	> 100	ND
bupropion hydroxylation	2B6	> 100	ND
amodiaquine N-deethylation	2C8	> 100	ND
diclofenac 4'-hydroxylation	2C9	> 100	ND
flurbiprofen 4'-hydroxylation	2C9	> 100	ND
S-mephenytoin 4'-hydroxylation	2C19	> 100	ND
dextromethorphan O-deethylation	2D6	> 100	ND
lauric acid 11-hydroxylation	2E1	> 100	ND
nifedipine oxidation	3A4	> 100	ND
testosterone 6ß-hydroxylation	3A4	> 100	ND
midazolam 1'-hydroxylation	3A4	> 100	ND
lauric acid 12-hydroxylation	4A11	> 100	ND

ND: not determined

Additional information:
no

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16.A PHARMACOKINETICS: OTHER

Test Article: Linagliptin

Location in Module 4: 4.2.2.5
Report No.: U06-3418
Study No.: PK0604T

Type of study: **In vitro elucidation of mechanism of renal secretion and reabsorption.**
Test item: **linagliptin**

Method: LLC-PK1 cells, an epithelial cell line derived from porcine kidney, was used to assess the renal proximal tubular seretion and reabsorption for organic cations. Bi-directional transcellular transport study was performed at 0.02, 0.2 and 0.2 µM linagliptin. Total transport clearance (sum of the uptake into cells and the transcellular transport) was compared between basal-to-apical (BtoA) and apical-to-basal (AtoB) directions in order to elucidate the mechanism of the nonlinear urinary excretion in human and the marked retention in rat kidney.

Results: Total transport clearance of linagliptin in the BtoA direction was higher than that in the AtoB direction at each tested concentration as well as TEA (tetraethylammonium), a typical compound that undergoes active urinary secretion.
In the BtoA direction, saturation of the uptake transport clearance and an increase of the transcellular transport clearance were observed as the concentration was increased from 0.02 µM to 0.2 µM linagliptin, though the total transport clearance was almost constant over this concentration range. These findings may suggest that saturable binding of linagliptin to an intracellular protein(s) that exhibits high affinity and low capacity characteristics occurs simulataneuously with renal secretion of linagliptin.

Tabulated results:

Substrate	(µM)	Transport clearance					
		Direction: BtoA			Direction: AtoB		
		CL_{total}	CL_{uptake}	$CL_{transcellular}$	CL_{total}	CL_{uptake}	$CL_{transcellular}$
TEA	9	20.9	3.50	17.4	1.56	0.684	0.878
L-carnitine	0.1	7.88	1.58	6.30	14.6	7.60	6.96
linagliptin	0.02	48.4	25.8	22.6	28.9	21.7	7.22
	0.2	51.9	8.31	43.6	13.2	5.17	8.06
	2	41.2	6.42	34.8	12.8	3.85	8.90

Unit for transport clearance: µL/5 min/cm^2 for TEA, µL/30 min/cm^2 for L-carnitine and linagliptin

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16.B PHARMACOKINETICS: OTHER

Test Article: Linagliptin

Location in Module 4: 4.2.2.3
Report No.: U06-3019
Study No.: PK05023

Type of study: **In vitro drug transporter profiling study. Active transport and inhibitory potential of linagliptin on drug transporters.**
Test item: **linagliptin**

Method: The presence of active transport of linagliptin by ten human SLC transporters was investigated by comparing the uptake of linagliptin in the transporter-expressing cells with that in the parental cells. The potential of linagliptin to inhibit 10 SLC transporter activities was assessed by comparing the uptake activity of a suitable probe substrate in the presence of linagliptin with that in the absence of linagliptin.
The presence of active transport of linagliptin by three human ABC transporters were investigated by comparing the degree of vectorial transport of linagliptin across the cell monolayer of the transporter-expressing cells with that of the parental cells. The potential of linagliptin to inhibit three ABC transporter activities was assessed by two approaches; e.g., transcellular and intravesicular transport. In the transcellular transport study using transwell, the effect of linagliptin was evaluated by comparing the degree of vectorial transport of a suitable probe substrate in the presence of linagliptin with that in the absence of linagliptin. Furthermore, in the intravesicular transport study using inside-out membrane vesicles expressing P-gp, MRP2 and BCRP, the inhibitory effect of linagliptin on the active intravesicular uptake of a suitable probe substrate was examined.
Following cell lines and membrane vesicles were used: OATP2-, OATP8-, OATP-B-, OCT1-, OCT2-, OCTN1-, OCTN2-, OAT1-, OAT3-, and OAT4-expressing HEK293, P-gp (MDR1)-expressing LLC-PK1 and MRP2- and BCRP-expressing cells, and membrane vesicles derived from P-gp-, MRP2- and BCRP-expressing MDCKII cells.

Results: linagliptin was a substrate of OATP8, OCT2, OAT4, OCTN1, OCTN2 and an inhibitor of OCT1 and OATP2. The IC_{50} values were 45.2 µM for OCT1 and 69.7 µM for OATP2. linagliptin is a substrate of P-gp and the Km value was 187 µM. linagliptin inhibited the P-gp activity with the IC_{50} value of 66.1 µM.

Tabulated results:

	Drug transporter isoform												
	SLC										ABC		
	OATP2	OATP8	OATP-B	OCT1	OCT2	OCTN1	OCTN2	OAT1	OAT3	OAT4	P-gp	MRP2	BCRP
Substrate	No	Yes	No	No	Yes	Yes	Yes	No	No	Yes	Yes	No	No
IC_{50} (µM)	69.7	>100	>100	45.2	>100	>100	>100	>100	>100	>100	66.1	>100	>100

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16.C PHARMACOKINETICS: OTHER

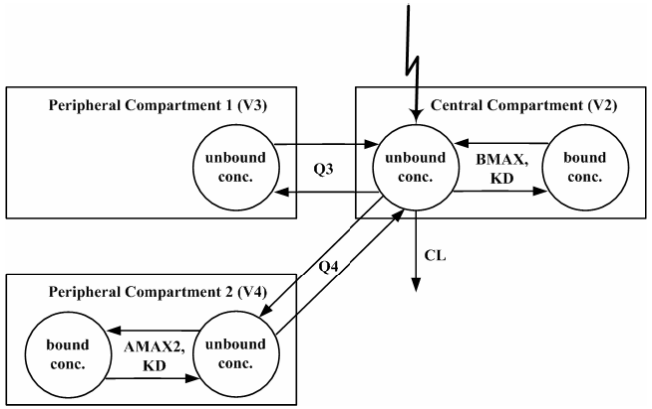
Test Article: Linagliptin

Location in Module 4: **4.2.2.3**
Report No.: **U09-1028**
Study No.: **A057/06FU**

Type of study: Development of a pharmacokinetic model that simultaneously describes the PK of linagliptin in DPP-4 deficient and in wild-type rats by accounting for saturable binding to DPP-4 only in the wild-type animals.

Method: Plasma levels of linagliptin after single intravenous (bolus) administration at various dose levels in DPP-4 deficient and wildtype Fischer rats were used from study U08-1240. NONMEM was used as software.

Tabulated results: A three compartment model was developed with two saturable binding sites, one in the central compartment and one in the peripheral compartment. This model described the pharmacokinetics of linagliptin in wildtype rats and was used for further simulations.



Parameter	Value	Description
V1 [L/kg]	2.28	Central volume of distribution
Q2 [(mL/min)/kg]	11.5	Inter-compartmental clearance between central compartment and peripheral compartment 1
V2 [L/kg]	6.92	Volume of distribution of the peripheral compartment 1
Q3 [(mL/min)/kg]	277	Inter-compartmental clearance between central compartment and peripheral compartment 2
V3 [L/kg]	7.24	Volume of distribution of the peripheral compartment 2
CL [mL/min/kg]	101	Clearance
BMAX [nmol/L]	2.70	Concentration of binding sites in the central compartment (just estimated for wild-type rats)
KON [L/(h ·nmol)]	2.50	KON rate for the non-linear binding (just estimated for wild-type rats)
AMAX2 [nmol/kg]	39.4	Amount of binding sites in the peripheral compartment 2 (just estimated for wild-type rats)
prop. residual variability [%]	19.9	Residual variability

Additional Information:
The model estimated DPP-4 binding parameters were consistent with the experimentally determined affinity constant, DPP-4 plasma concentration and total amount of peripheral DPP-4. Based on the PK model an intravenous dose of 0.04 mg/kg linagliptin is needed to saturate >95% of the peripheral DPP-4 molecules in rats.